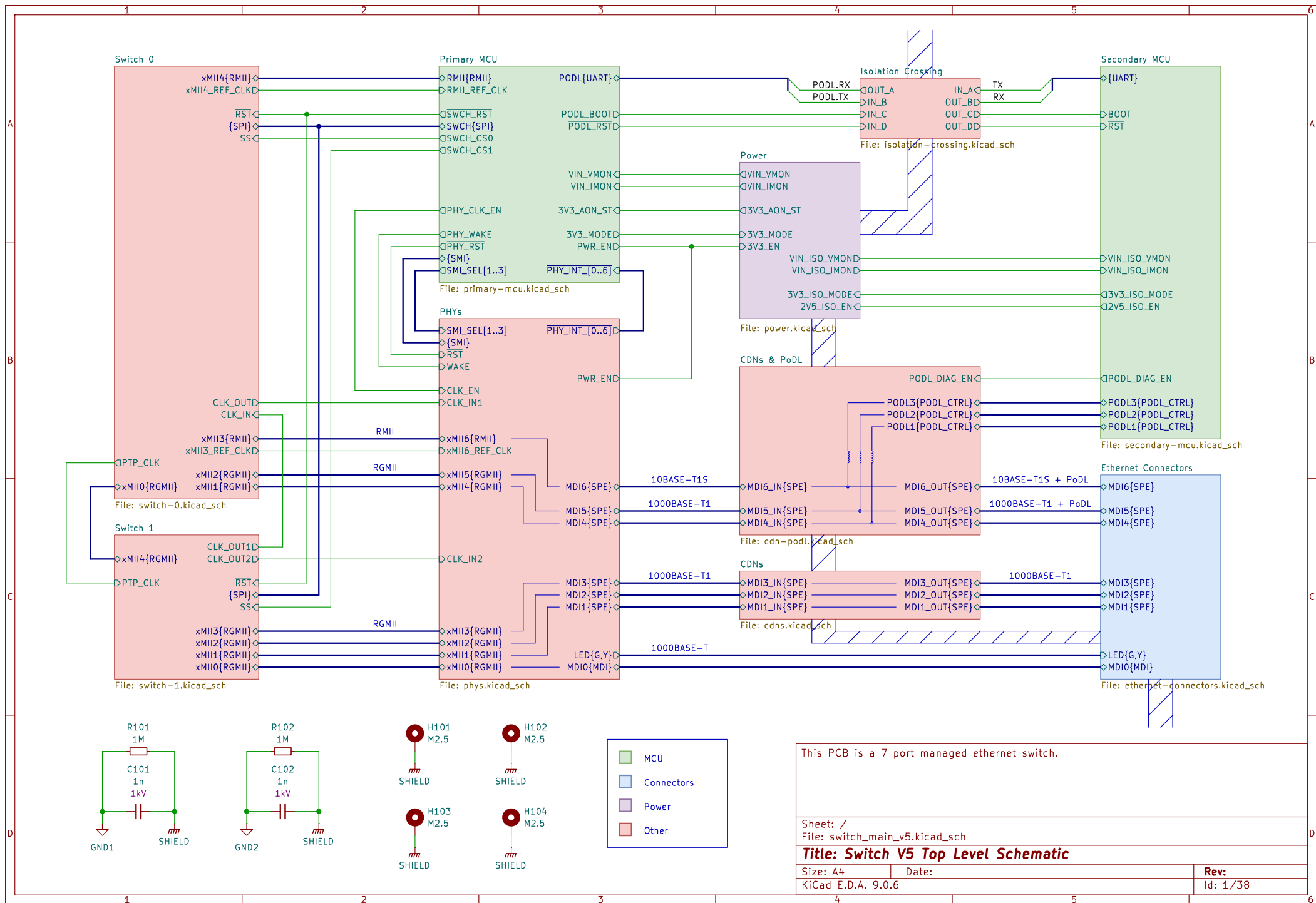
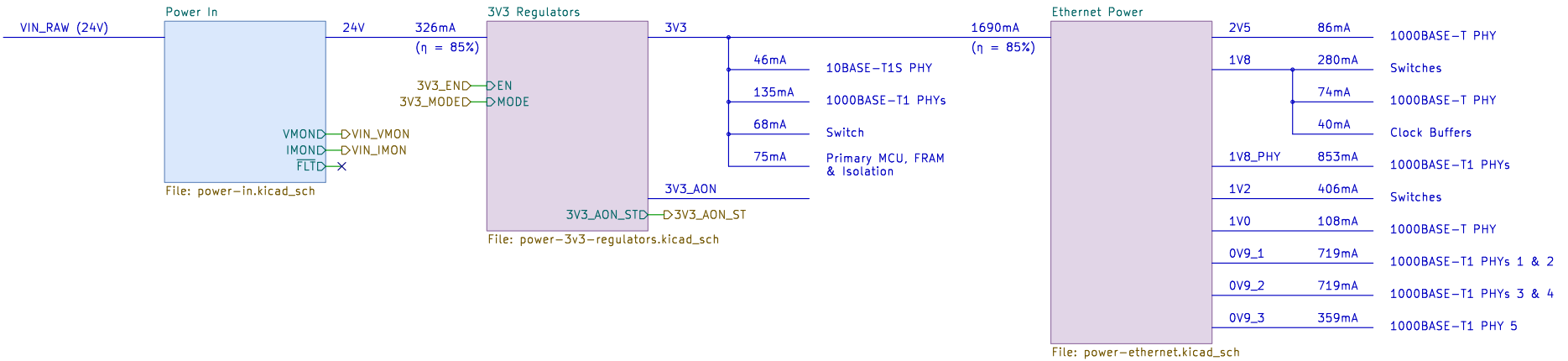


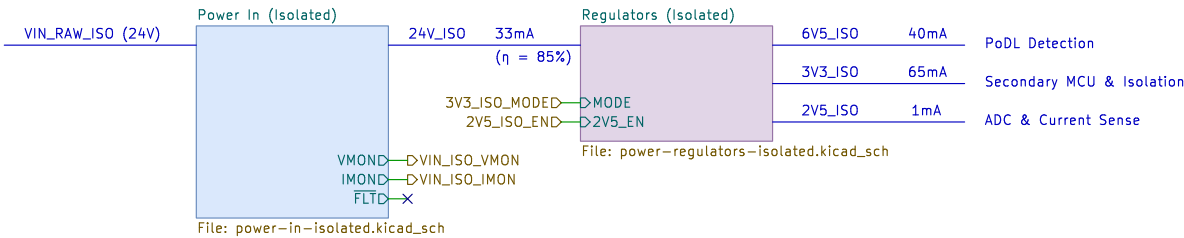


PHY Side Power



Isolation Barrier

Connector Side Power



Block diagram of the power supply scheme for both sides of the isolation barrier.

Sheet: /Power/
File: power.kicad_sch

Title: Power

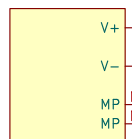
Size: A4
KiCad E.D.A. 9.0.6

Date:

Rev:

Id: 2/38

J301
MKXS1-1 Power Input



ESD
Protection

Surge
Protection

VIN_RAW

D301
SZESD7241N2T5G



U302
TVS3301DRBR

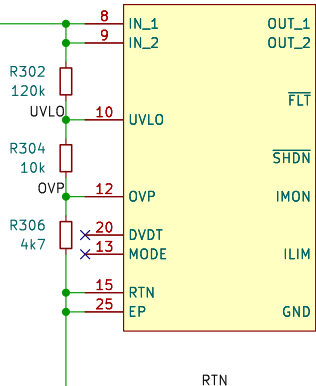
C301
100n
100V

GND1

UVLO = 10.9V
OVP = 34.1V

eFuse

U301
TPS26600RHFR



$VMON = VIN / 18.857$
The 150mΩ eFuse
resistance should be
taken into account when
calculating input power.

+VIN

R301
100k
VMON
R303
5k6
GND1

$I_{OUT} = VMON / 7.84$

$R_{ILIM} = 22k\Omega$
 $\therefore I_{LIM} = 12 / 22 = 545mA$

TPS26600 has auto-retry
when a fault occurs.

Power input and protection for the PHY side of the PCB.

Sheet: /Power/Power In/
File: power-in.kicad_sch

Title: Power In (PHY Side)

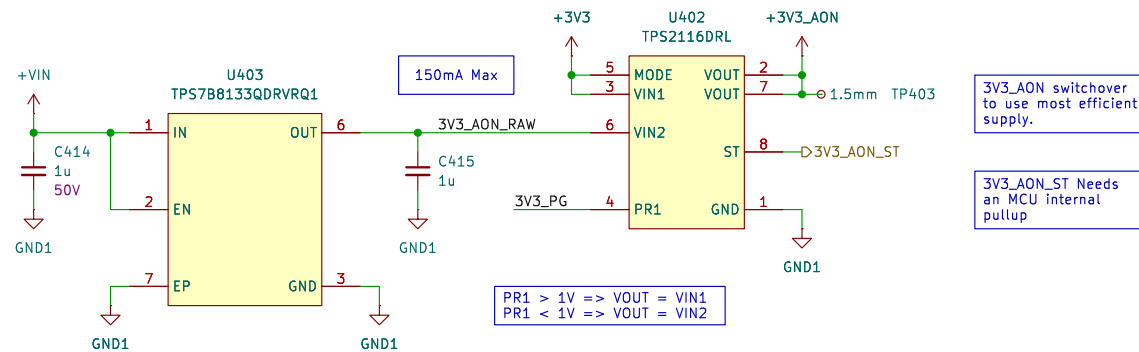
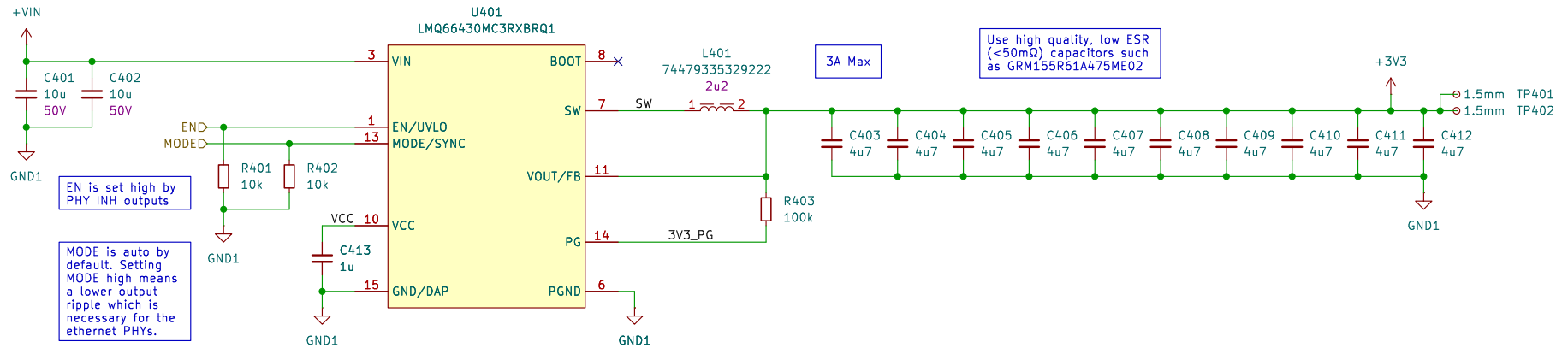
Size: A4

Date:

KiCad E.D.A. 9.0.6

Rev:

Id: 3/38



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Id: 4/38



Sheet: /Power/Ethernet Power/
File: power-ethernet.kicad_sch

Title: Ethernet Power (PHY Side)

Size: A4

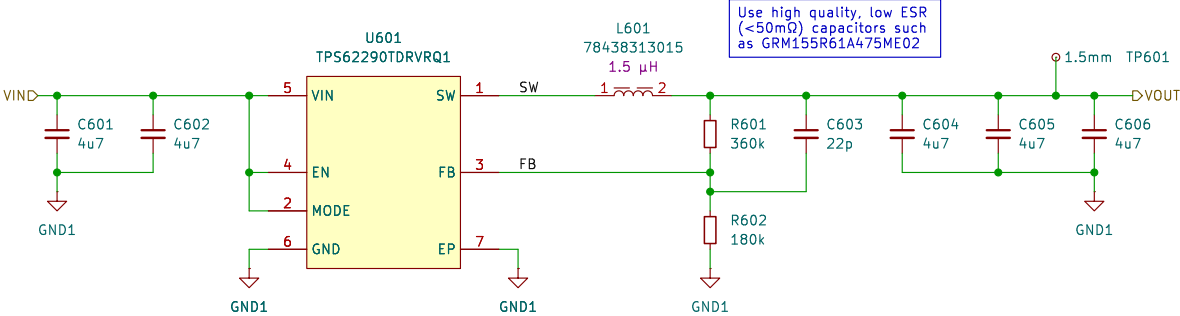
Date:

KiCad E.D.A. 9.0.6

Rev:

Id: 5/38

1V8 Regulator



1.8V, 1A buck SMPS.

Sheet: /Power/Ethernet Power/1V8 Buck Converter/
File: 1v8_buck_converter.kicad_sch

Title:

Size: A4

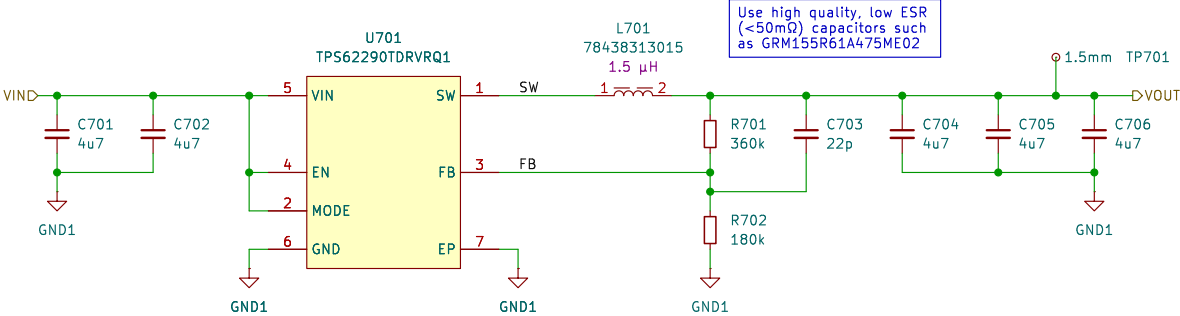
Date:

KiCad E.D.A. 9.0.6

Rev:

Id: 6/38

1V8 Regulator



1.8V, 1A buck SMPS.

Sheet: /Power/Ethernet Power/1V8_PHY Buck Converter/
File: 1v8_buck_converter.kicad_sch

Title:

Size: A4

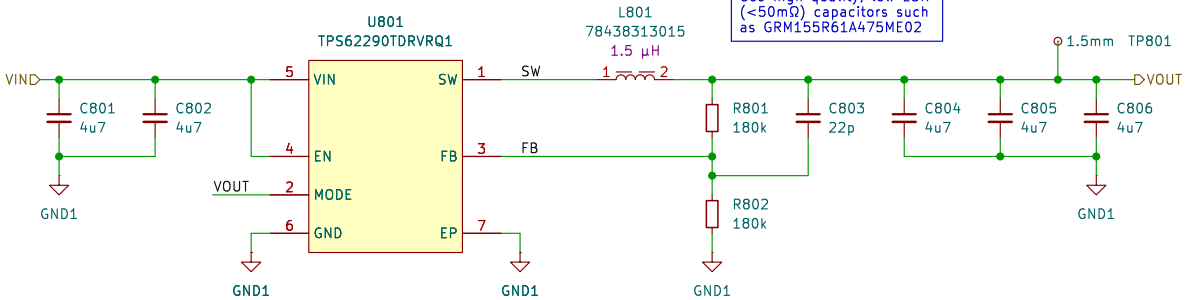
Date:

KiCad E.D.A. 9.0.6

Rev:

Id: 7/38

1V2 Regulator



Use high quality, low ESR (<50mΩ) capacitors such as GRM155R61A475ME02

1.2V, 1A buck SMPS.

Sheet: /Power/Ethernet Power/1V2 Buck Converter/
File: 1v2_buck_converter.kicad_sch

Title:

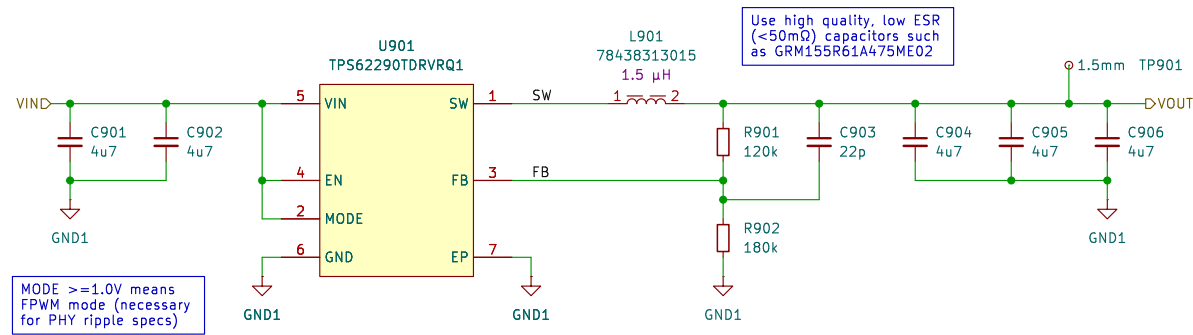
Size: A4

Date:

Rev:

Id: 8/38

1V0 Regulator



1.0V, 1A buck SMPS.

Sheet: /Power/Ethernet Power/1V0 Buck Converter/
File: 1v0_buck_converter.kicad_sch

Title:

Size: A4

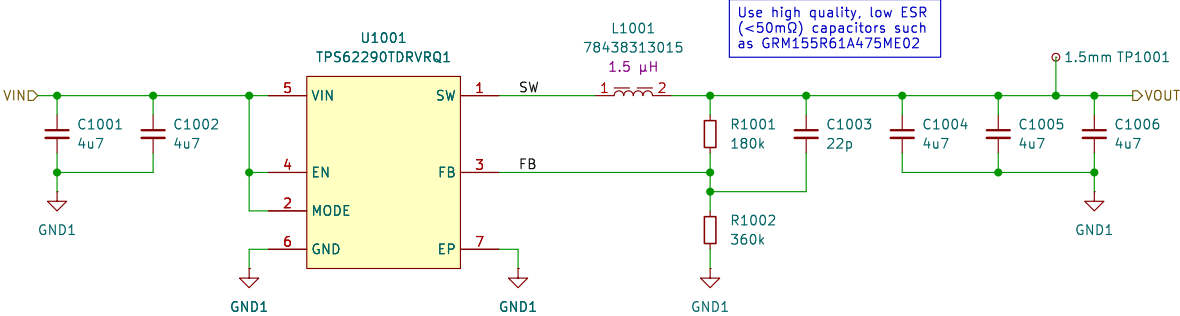
Date:

KiCad E.D.A. 9.0.6

Rev:

Id: 9/38

0V9 Regulator



Use high quality, low ESR (<50mΩ) capacitors such as GRM155R61A475ME02

0.9V, 1A buck SMPS.

Sheet: /Power/Ethernet Power/0V9_1 Buck Converter/
File: 0v9_buck_converter.kicad_sch

Title:

Size: A4

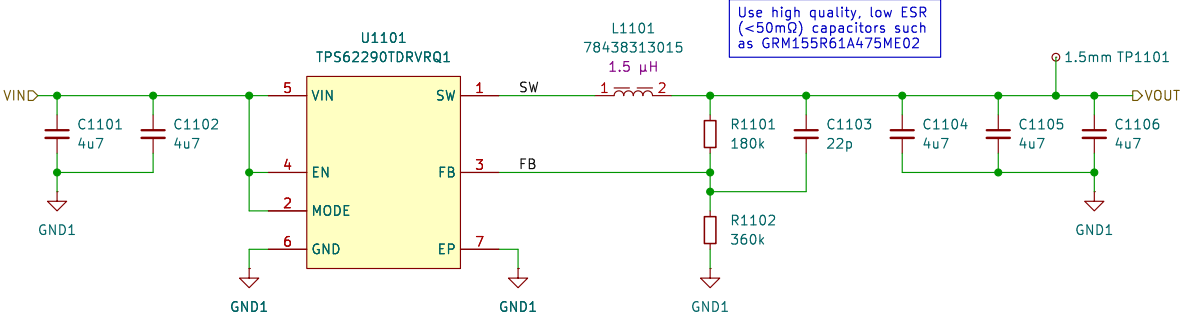
Date:

KiCad E.D.A. 9.0.6

Rev:

Id: 10/38

0V9 Regulator



0.9V, 1A buck SMPS.

Sheet: /Power/Ethernet Power/0V9_2 Buck Converter/
File: 0v9_buck_converter.kicad_sch

Title:

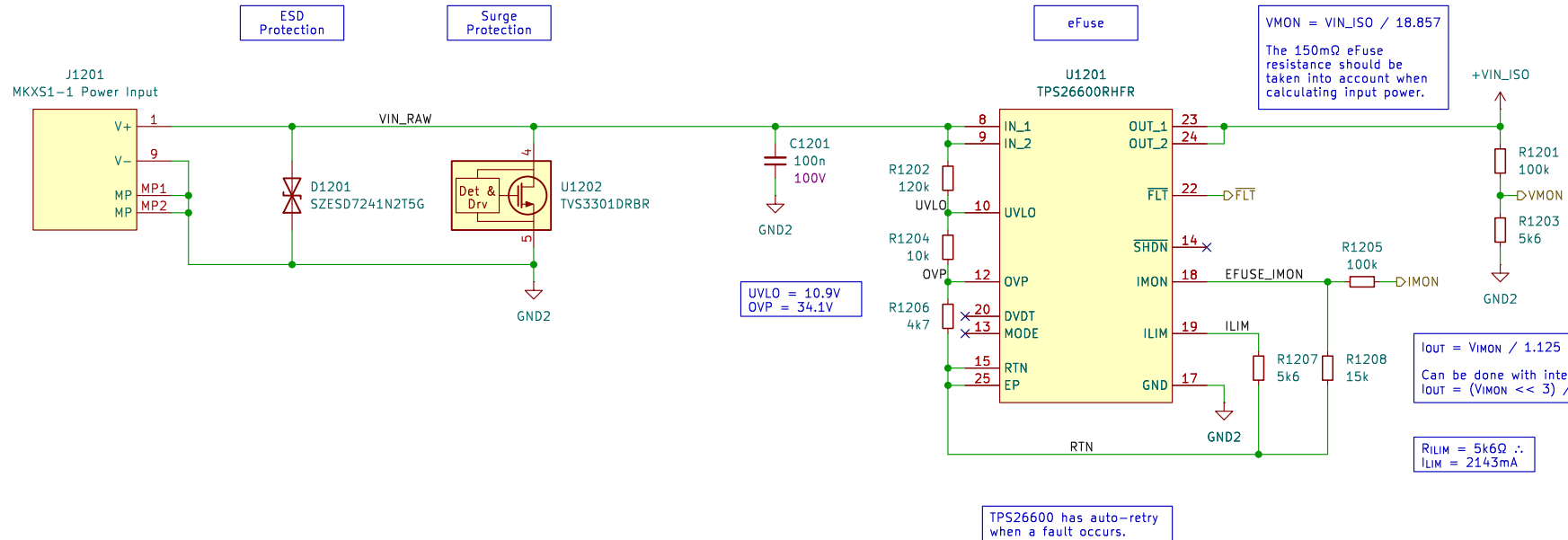
Size: A4

Date:

KiCad E.D.A. 9.0.6

Rev:

Id: 11/38



Power input and protection for the connector side of the PCB.

Sheet: /Power/Power In (Isolated)/
File: power-in-isolated.kicad_sch

Title: Power In (Connector Side)

Size: A4

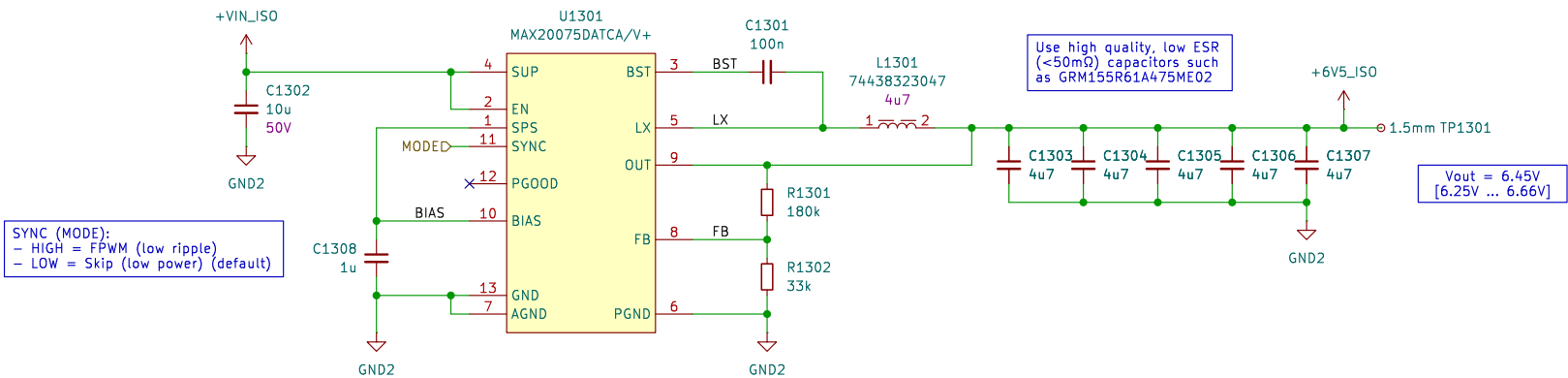
Date:

Rev:

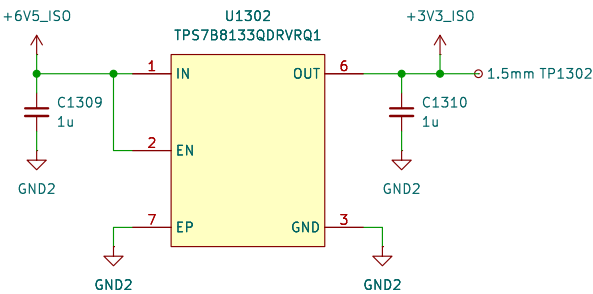
KiCad E.D.A. 9.0.6

Id: 12/38

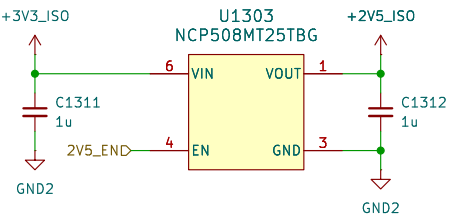
6V5 Regulator



3V3 Regulator



2V5 Regulator



6V5, 3V3 and 2V5 regulators for the connector side of the PCB.

Sheet: /Power/Regulators (Isolated)/
File: power-regulators-isolated.kicad_sch

Title: Regulators (Connector Side)

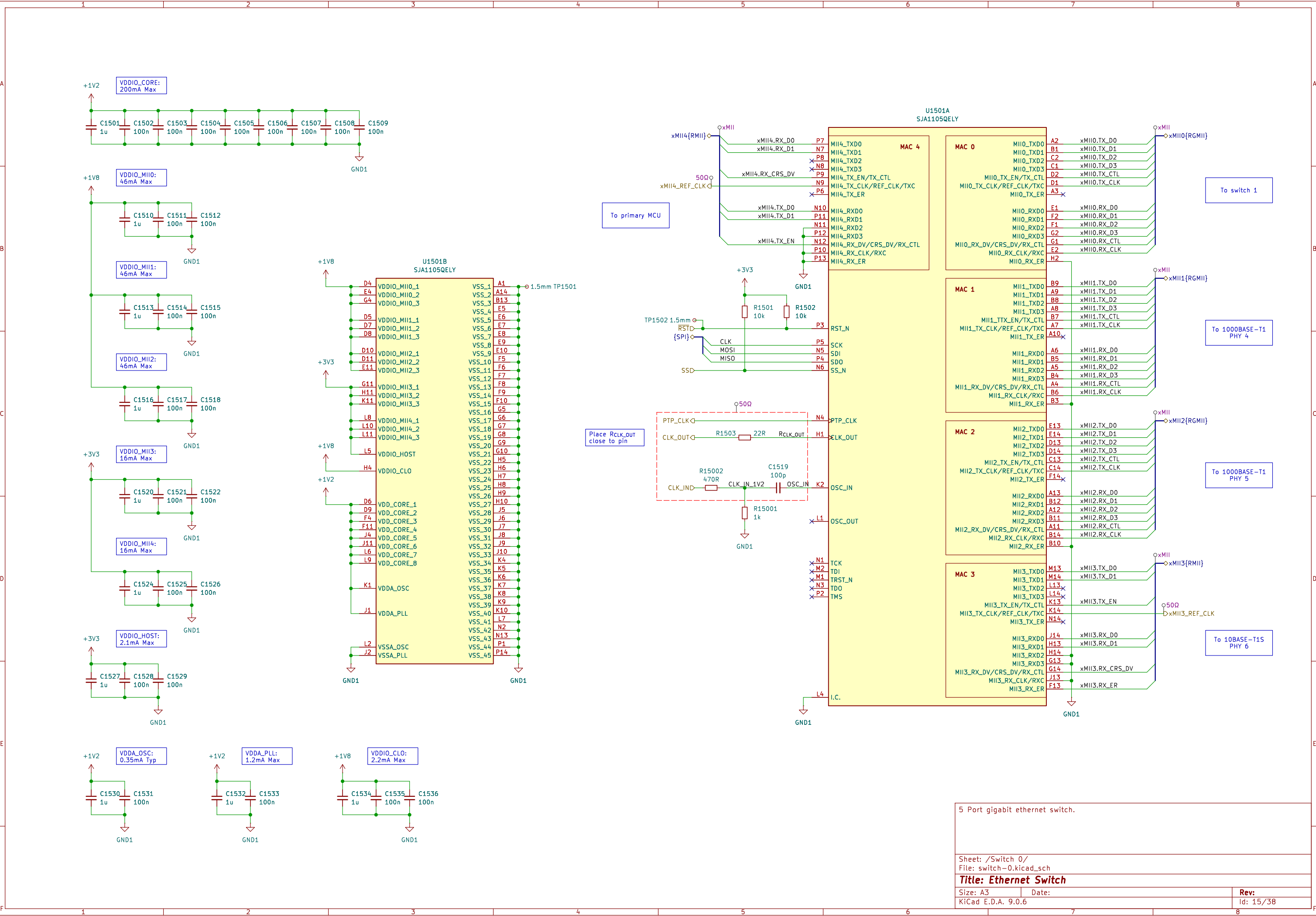
Size: A4

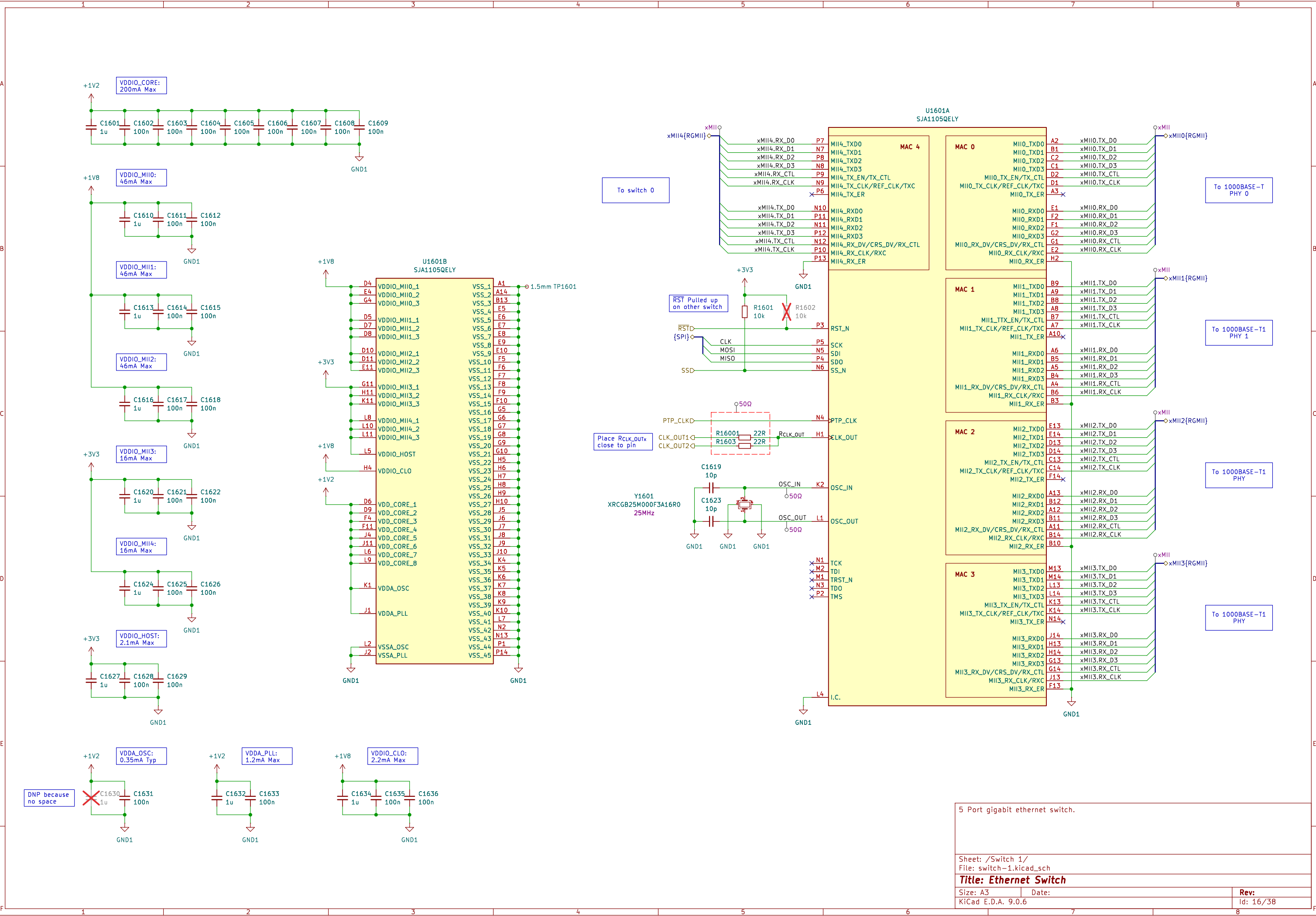
Date:

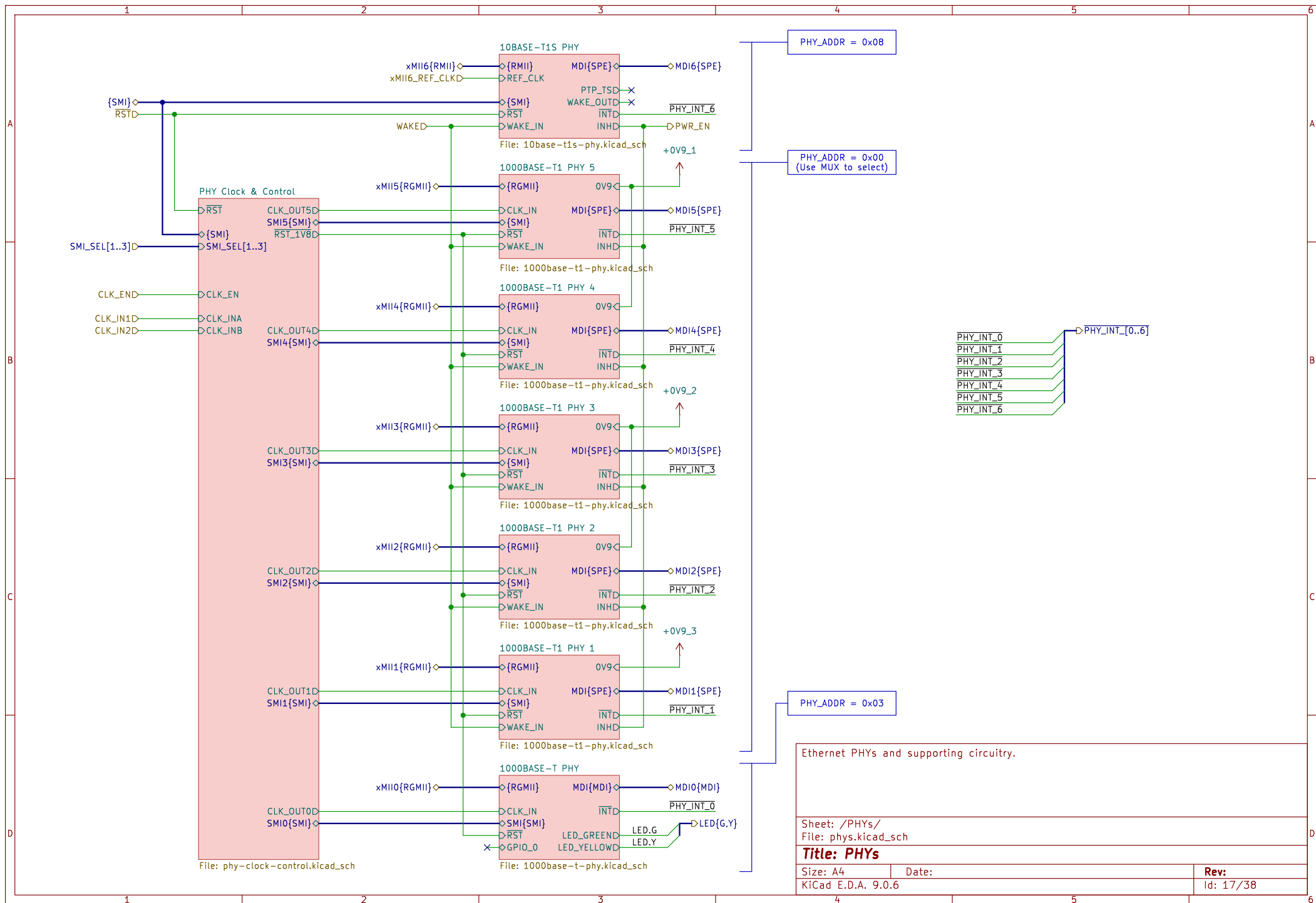
Rev:

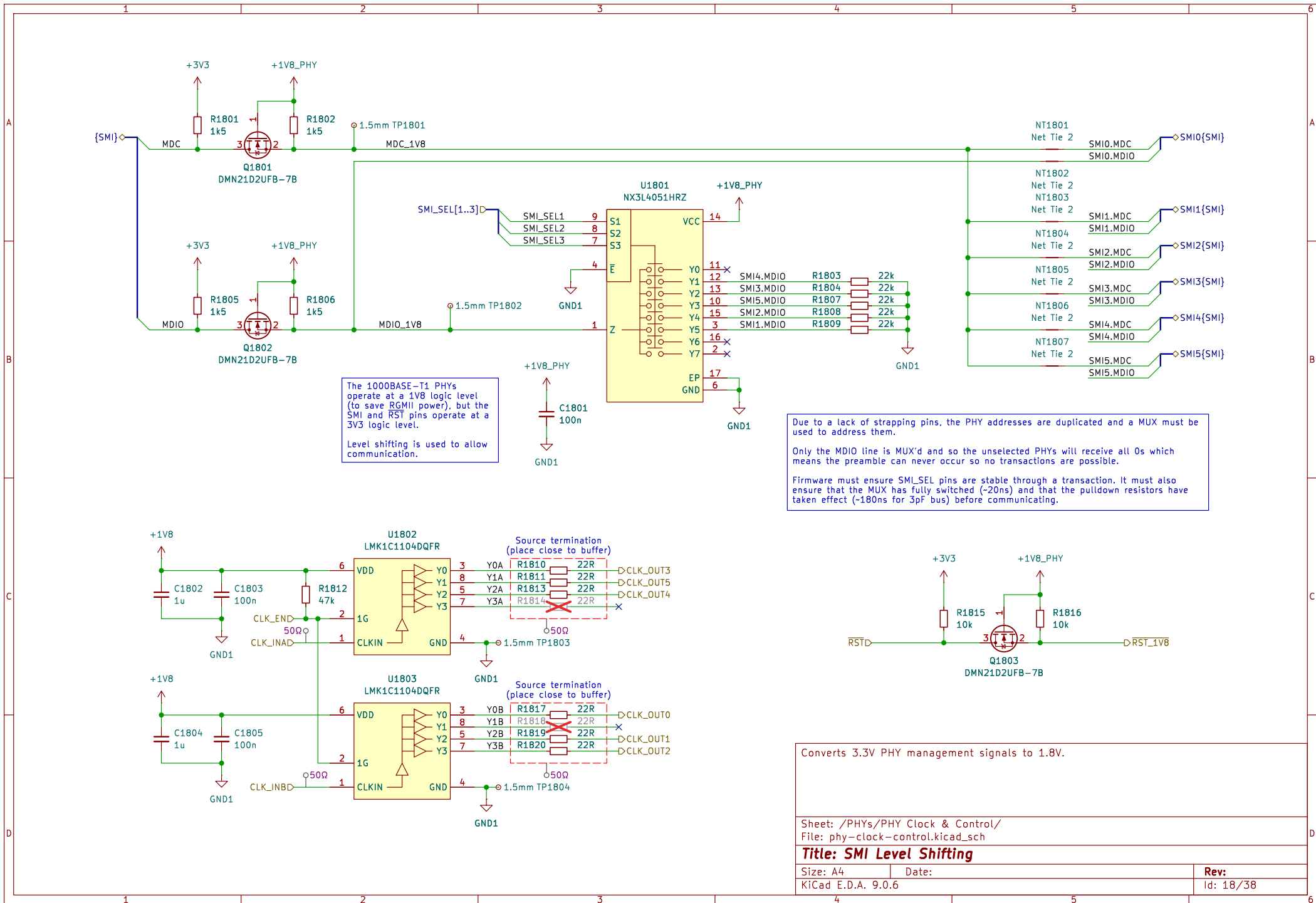
KiCad E.D.A. 9.0.6

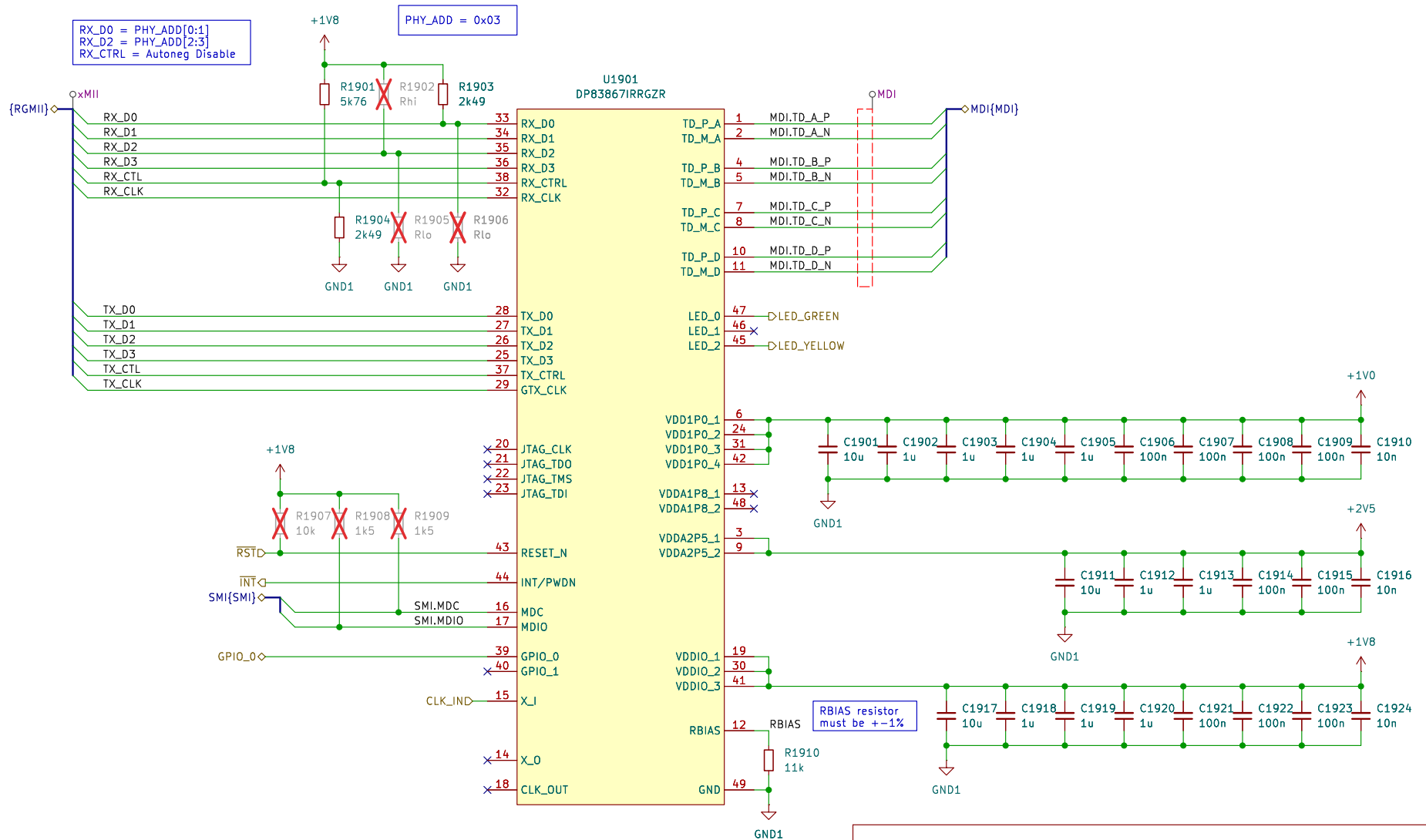
Id: 13/38











Sheet: /PHYs/1000BASE-T PHY/
File: 1000base-t-phy.kicad_sch

Title: PHY

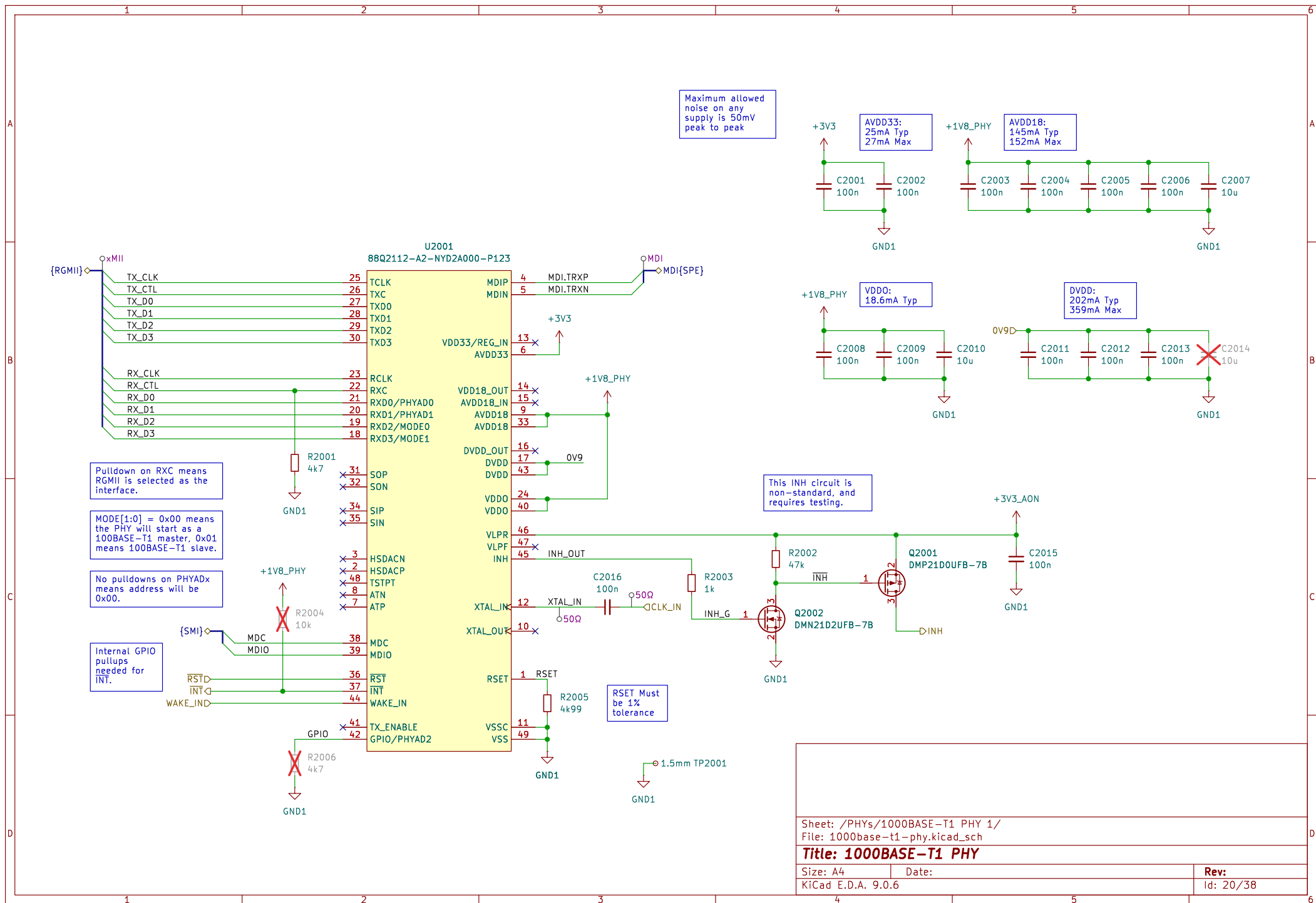
Size: A4

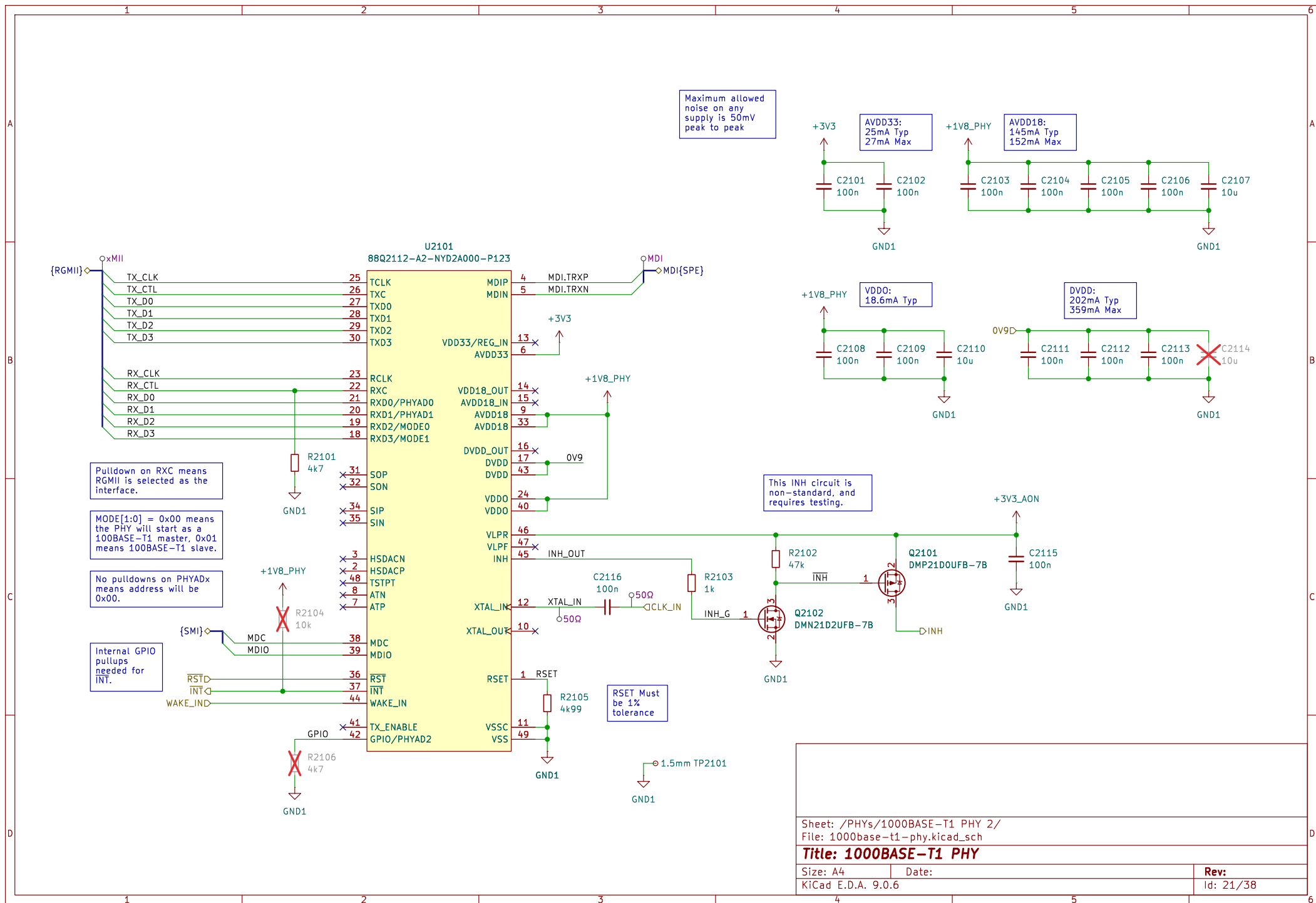
Date:

KiCad E.D.A. 9.0.6

Rev: 1.0.0

Id: 19/38





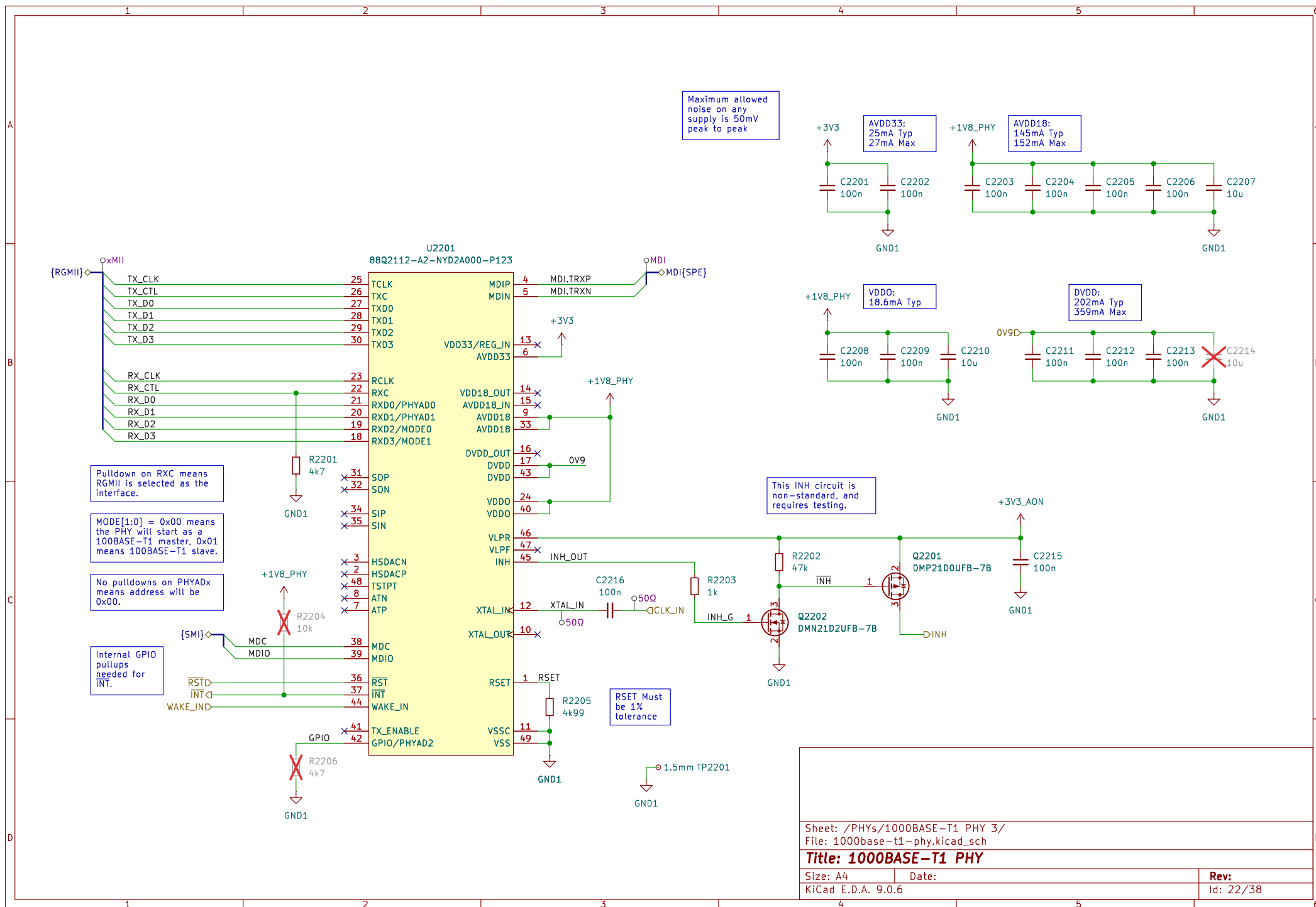
Sheet: /PHYs/1000BASE-T1 PHY 2/
File: 1000base-t1-phy.kicad_sch

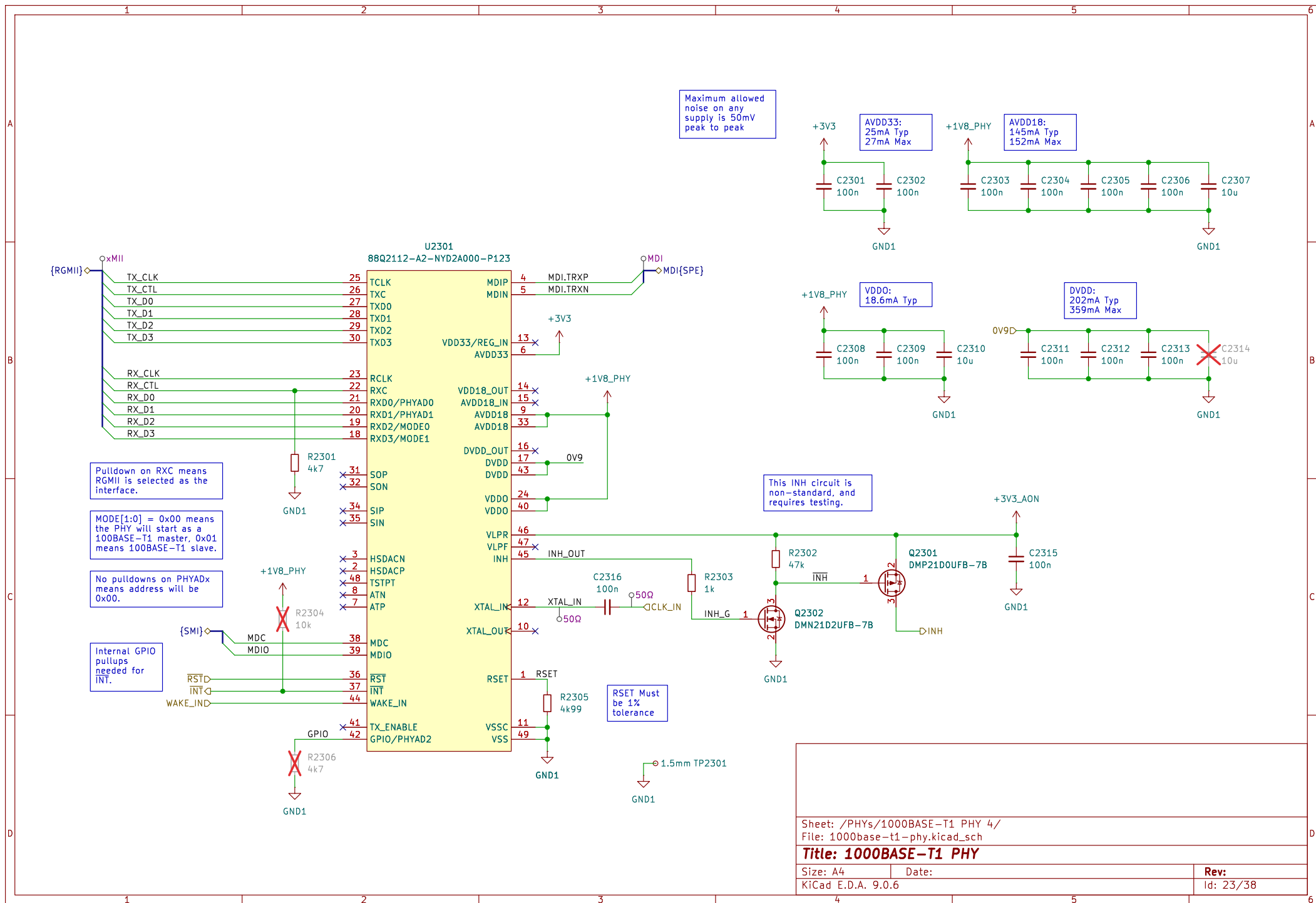
Title: 1000BASE-T1 PHY

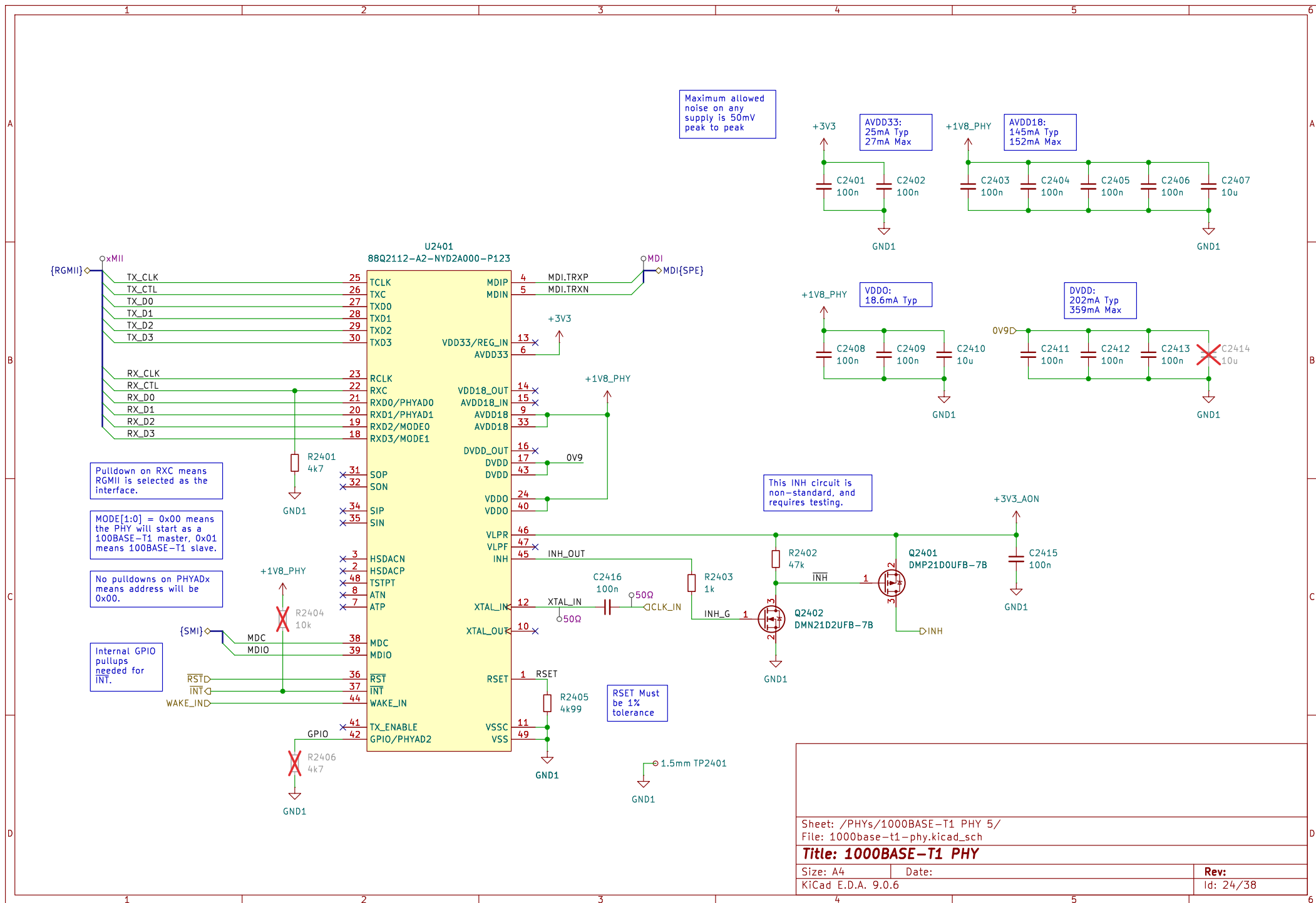
Size: A4
KiCad E.D.A. 9.0.6

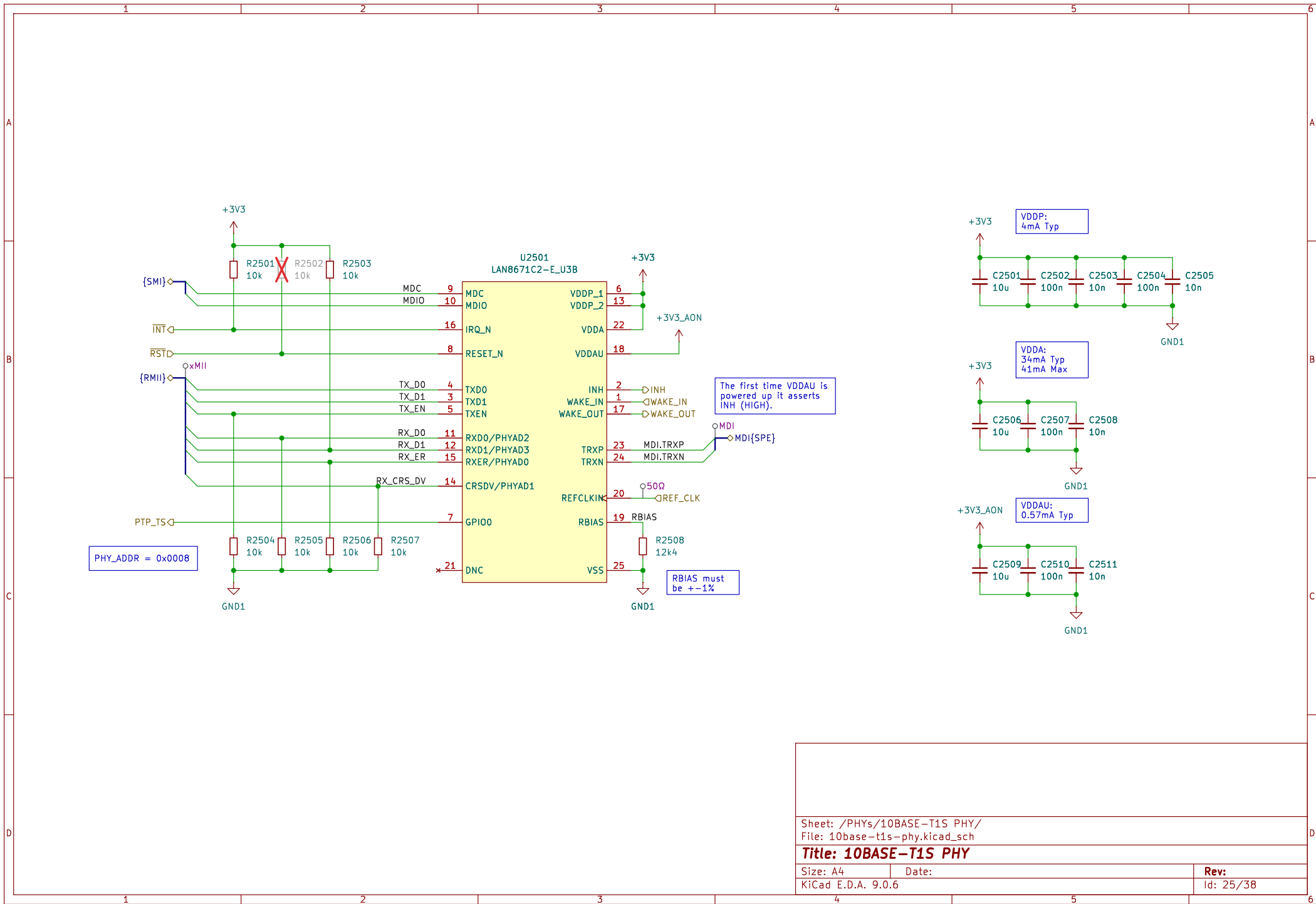
Date:

Rev:
Id: 21/38

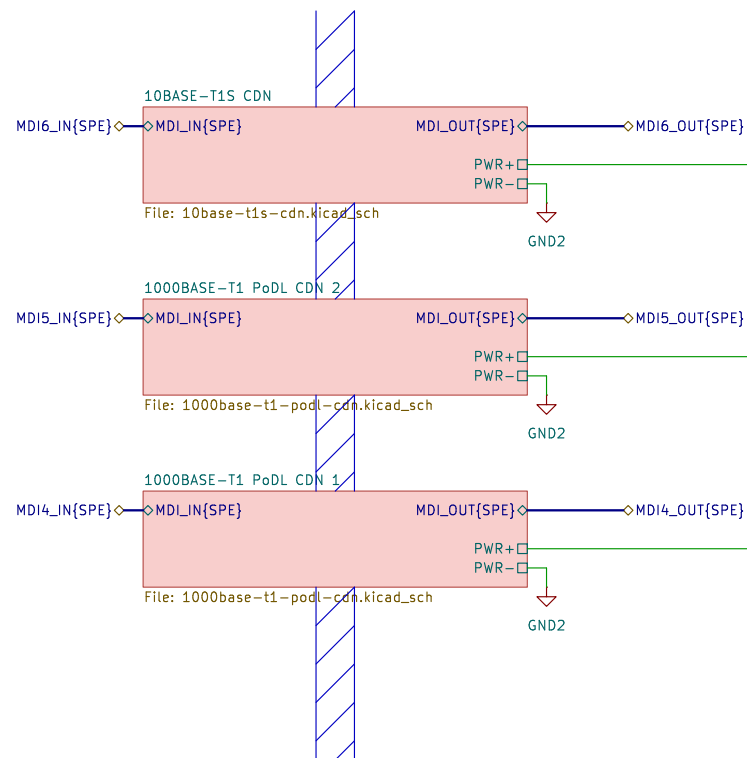




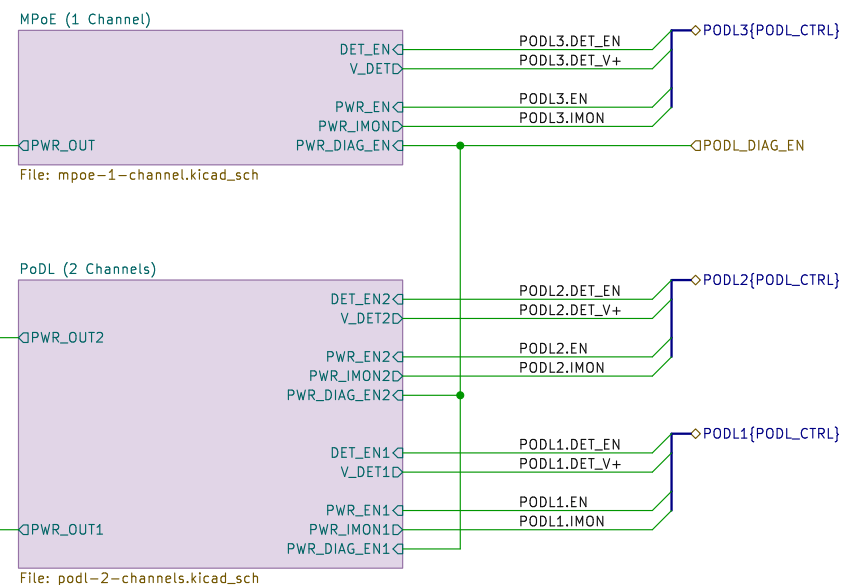




Coupling-Decoupling Networks (CDNs)



Power Injection



Coupling-decoupling networks (CDNs) and power over datalines (PoDL).

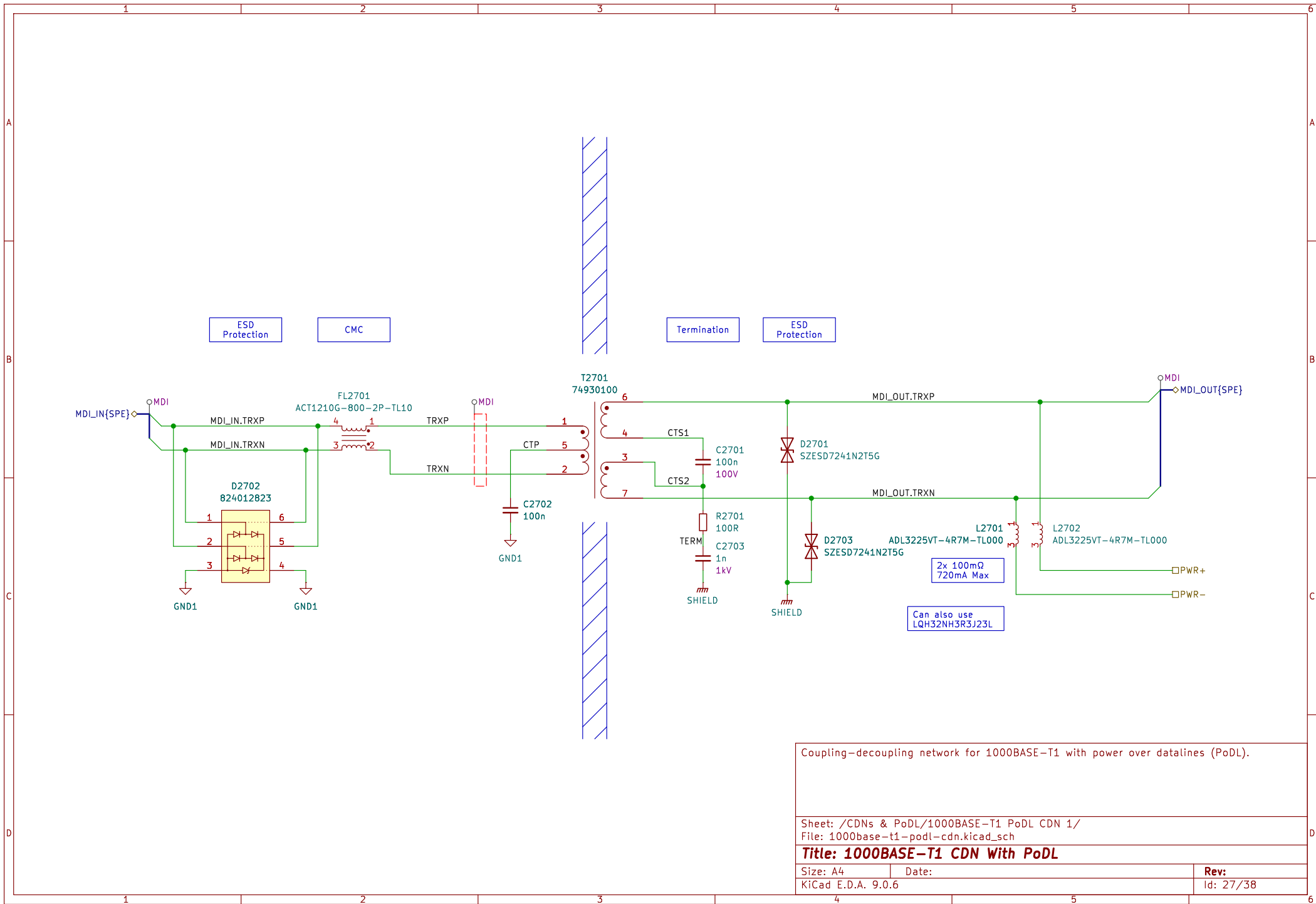
Sheet: /CDNs & PoDL/
File: cdn-podl.kicad_sch

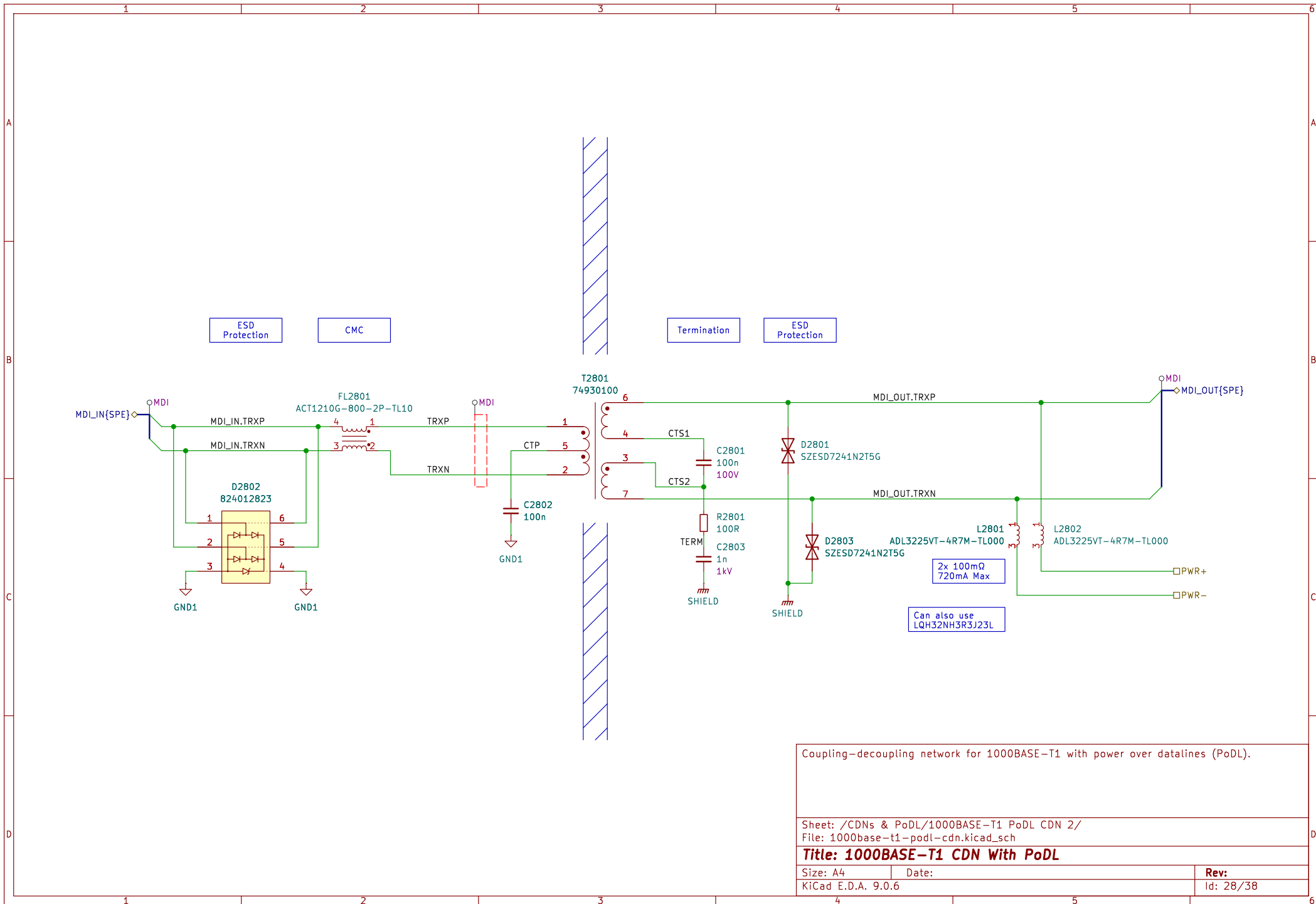
Title: CDNs and PoDL

Size: A4
KiCad E.D.A. 9.0.6

Date:

Rev:
Id: 26/38



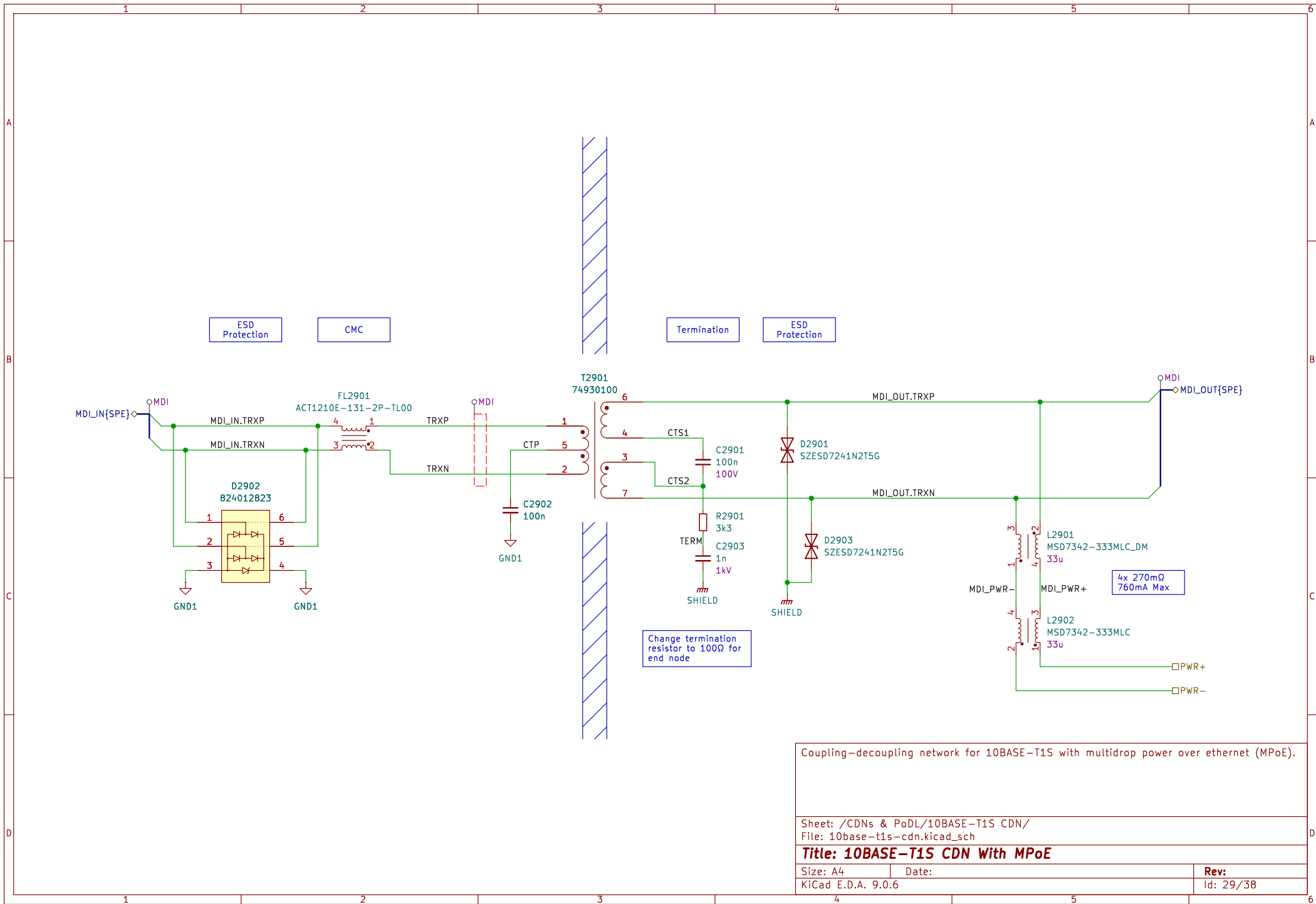


Coupling-decoupling network for 1000BASE-T1 with power over datalines (PoDL).

Sheet: /CDNs & PoDL/1000BASE-T1 PoDL CDN 2/
File: 1000base-t1-podl-cdn.kicad_sch

Title: 1000BASE-T1 CDN With PoDL

Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.6		Id: 28/38



Coupling-decoupling network for 10BASE-T1S with multidrop power over ethernet (MPoE).

Sheet: /CDNs & PoDL/10BASE-T1S CDN/
File: 10base-t1s-cdn.kicad_sch

Title: 10BASE-T1S CDN With MPoE

Size: A4

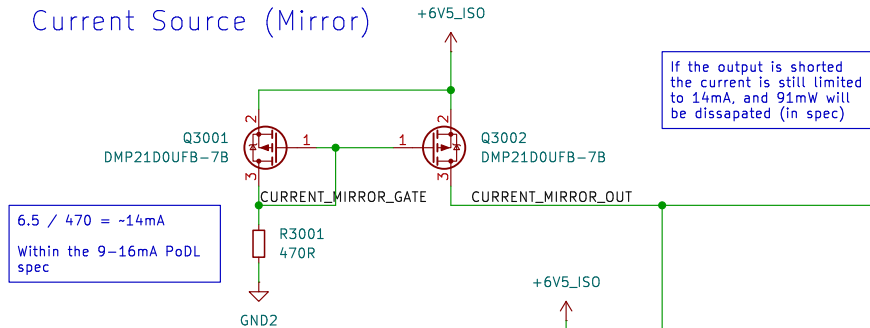
Date:

KiCad E.D.A. 9.0.6

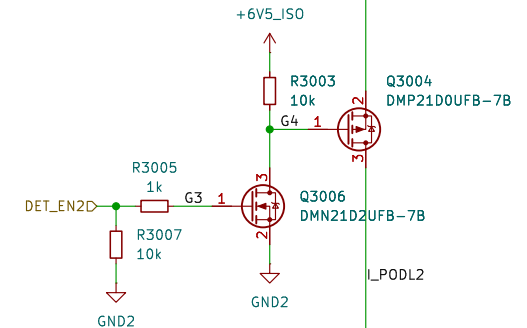
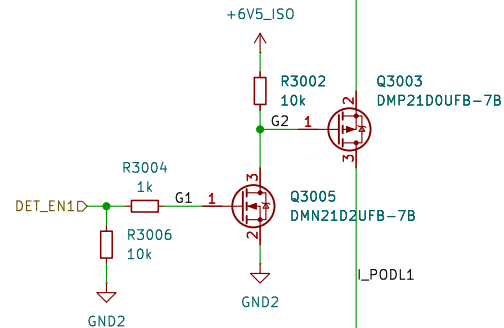
Rev:

Id: 29/38

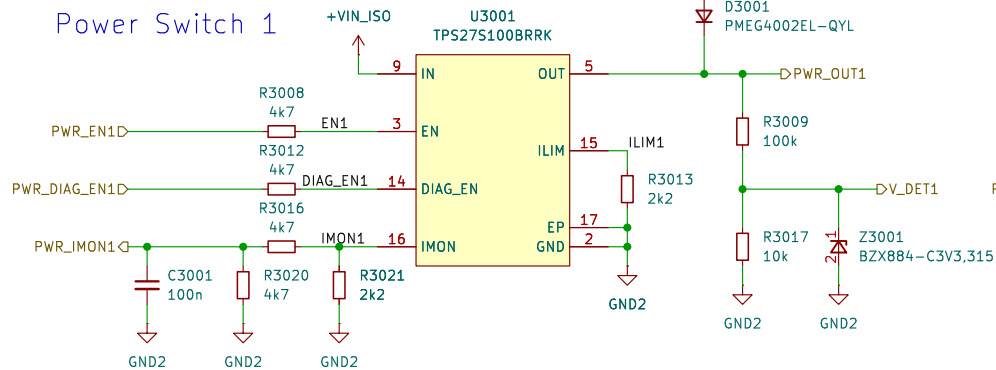
Current Source (Mirror)



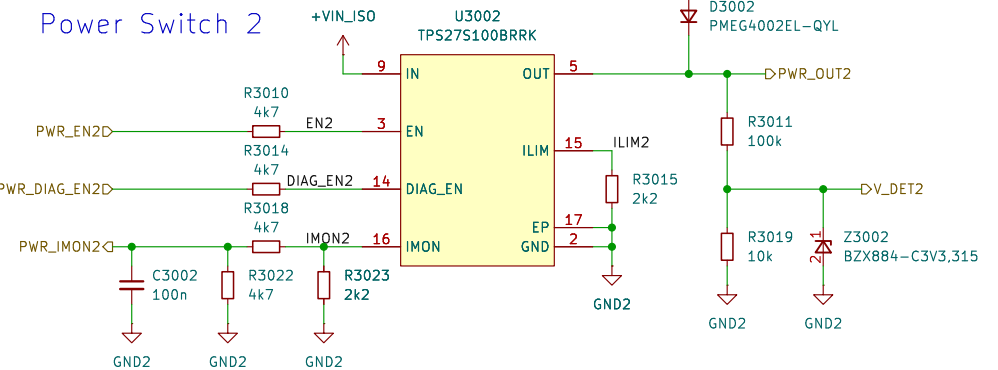
Current Switches



Power Switch 1



Power Switch 2



1.1A Maximum output current
1.1A Maximum measurable current

IMON can go up to 4.75V, potential divider limits this to be within 2.5V ADC range.

1/11 Potential divider and zener clamp to protect the 3V3 microcontroller IO.

$$I_{OUT} = 0.28046 \cdot V_{MON}$$

During the detection phase, 9-16mA is sent into TRXP and the resulting voltage is measured.
If a PoDL compatible device is detected on the other end (indicated by a 4.05-4.55V zener diode) then the full 24V is enabled.

Sheet: /CDNs & PoDL/PoDL (2 Channels)/
File: podl-2-channels.kicad_sch

Title: PoDL Detection, Switching and Monitoring

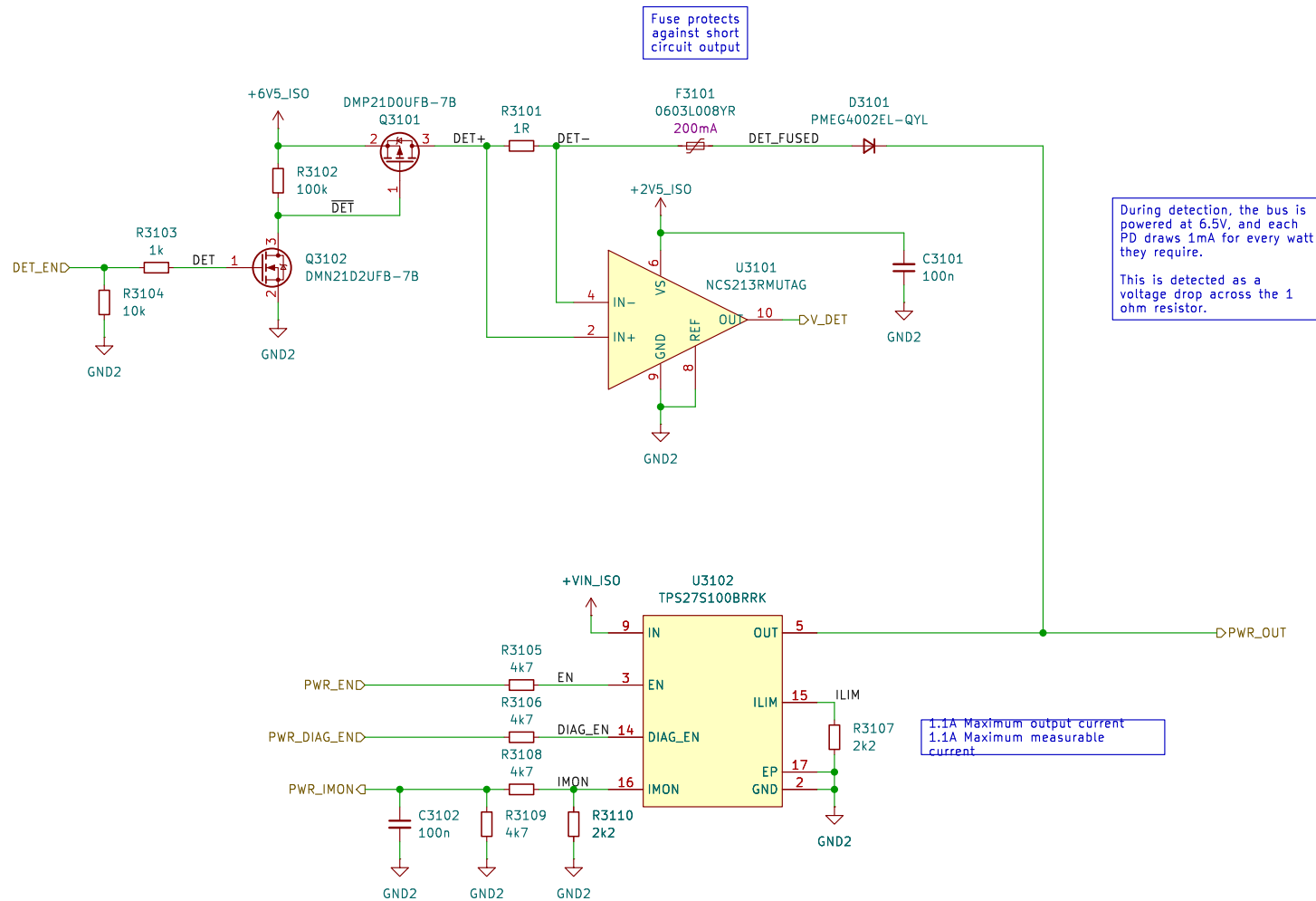
Size: A4

Date:

Rev:

KiCad E.D.A. 9.0.6

Id: 30/38



Sheet: /CDNs & PoDL/MPoE (1 Channel)/
File: mpoe-1-channel.kicad_sch

Title: MPoE Detection, Switching and Monitoring

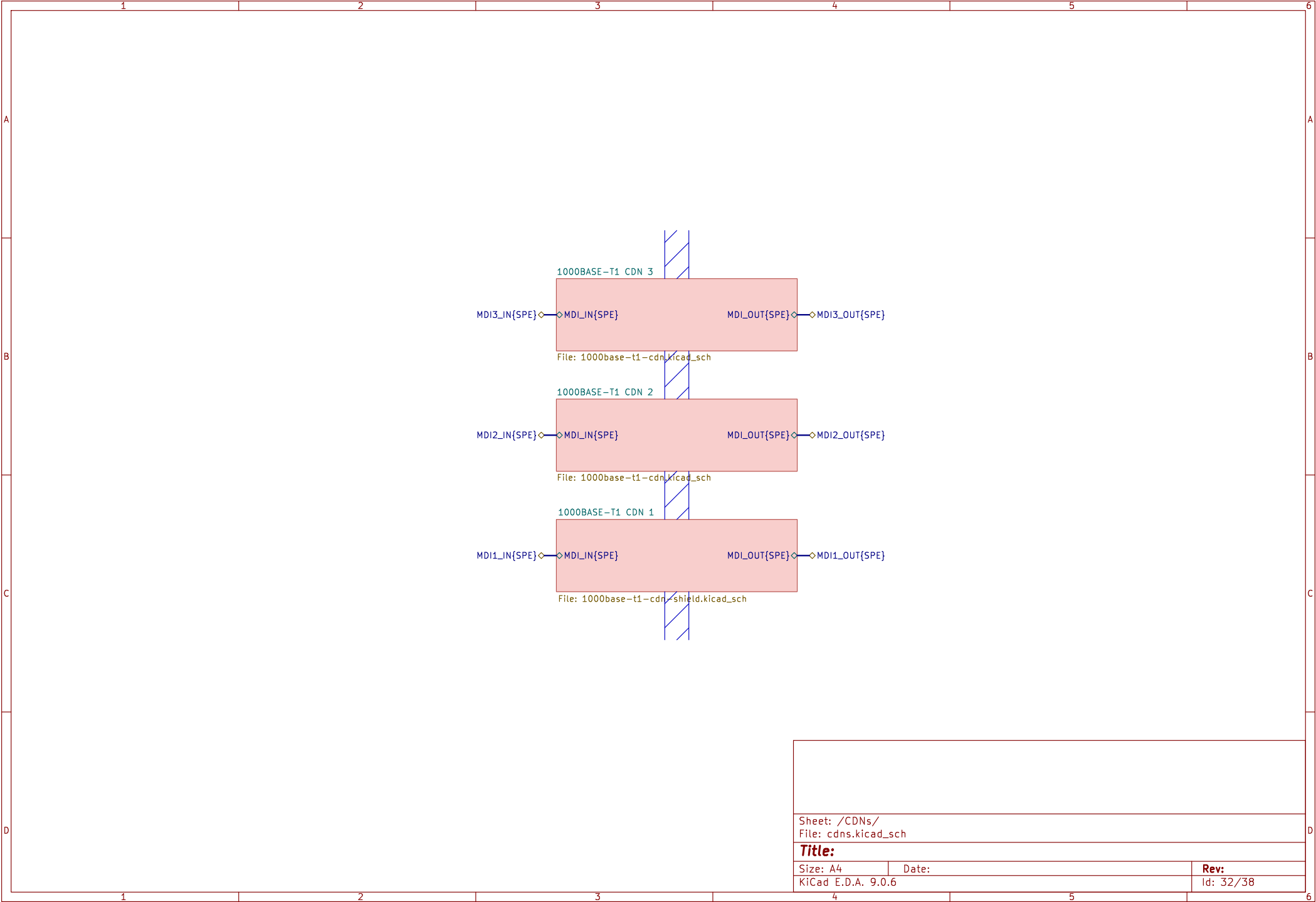
Size: A4

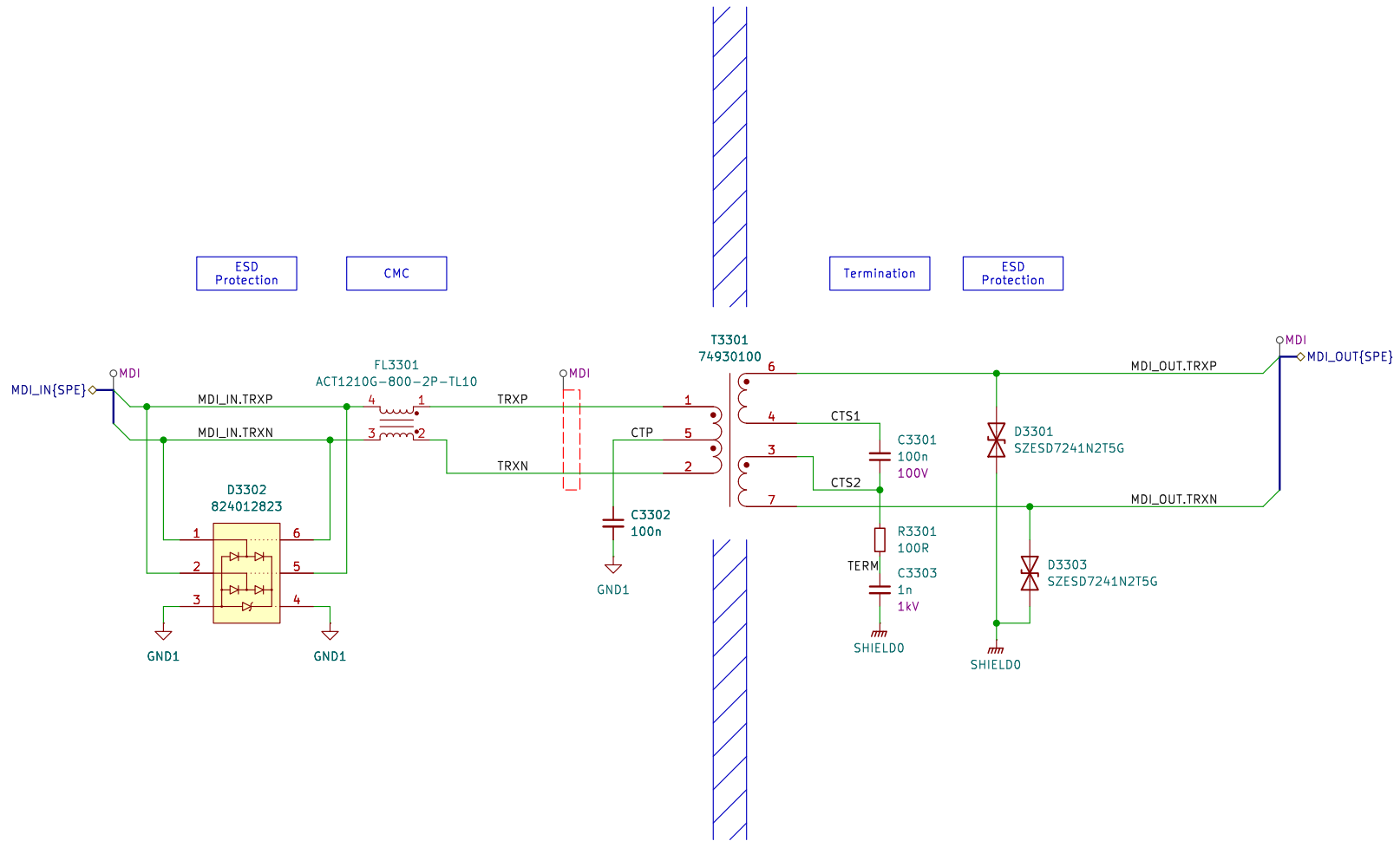
Date:

Rev:

KiCad E.D.A. 9.0.6

Id: 31/38





Coupling-decoupling network for 1000BASE-T1

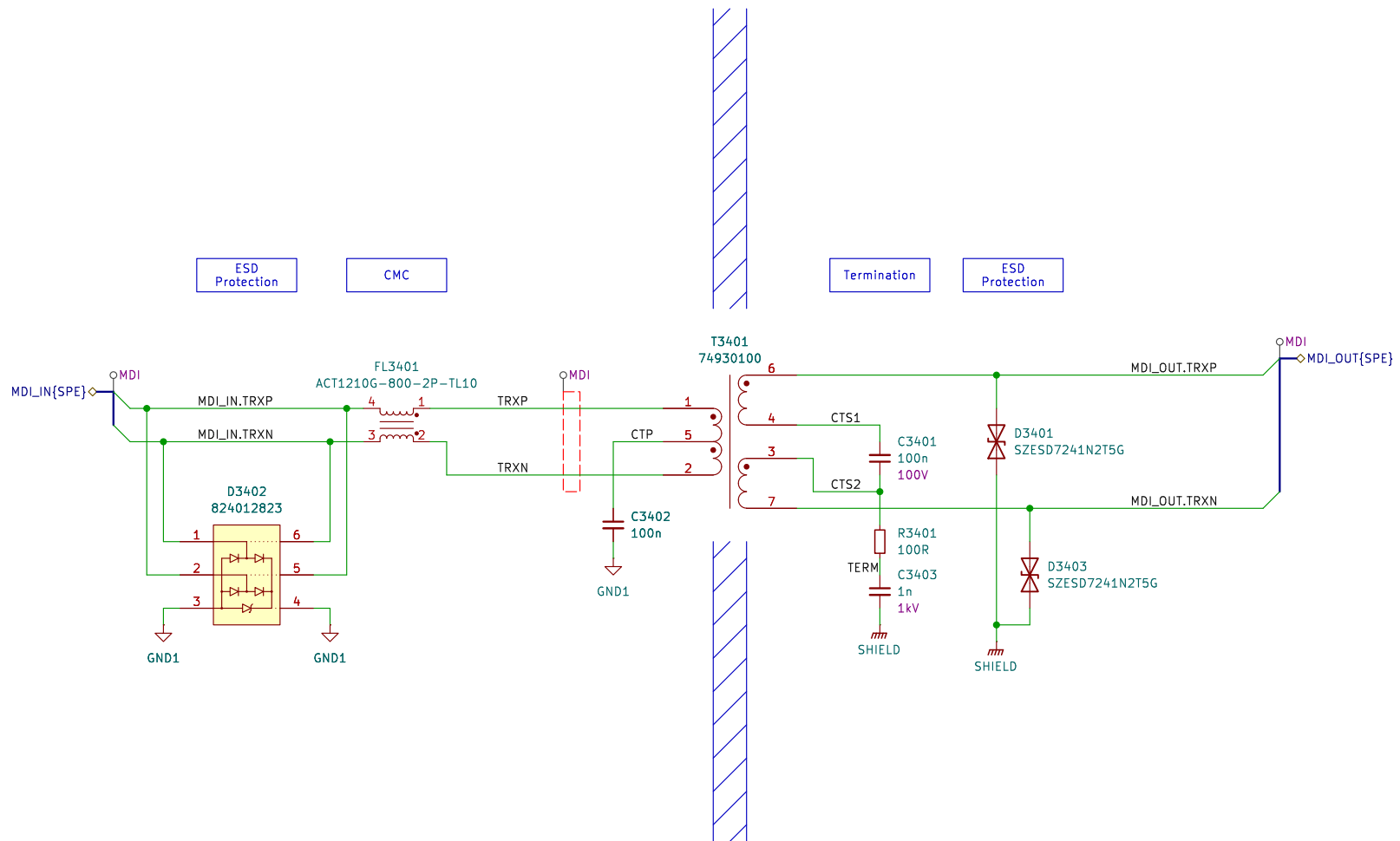
Sheet: /CDNs/1000BASE-T1 CDN 1/
File: 1000base-t1-cdn-shield.kicad_sch

Title: 1000BASE-T1 CDN

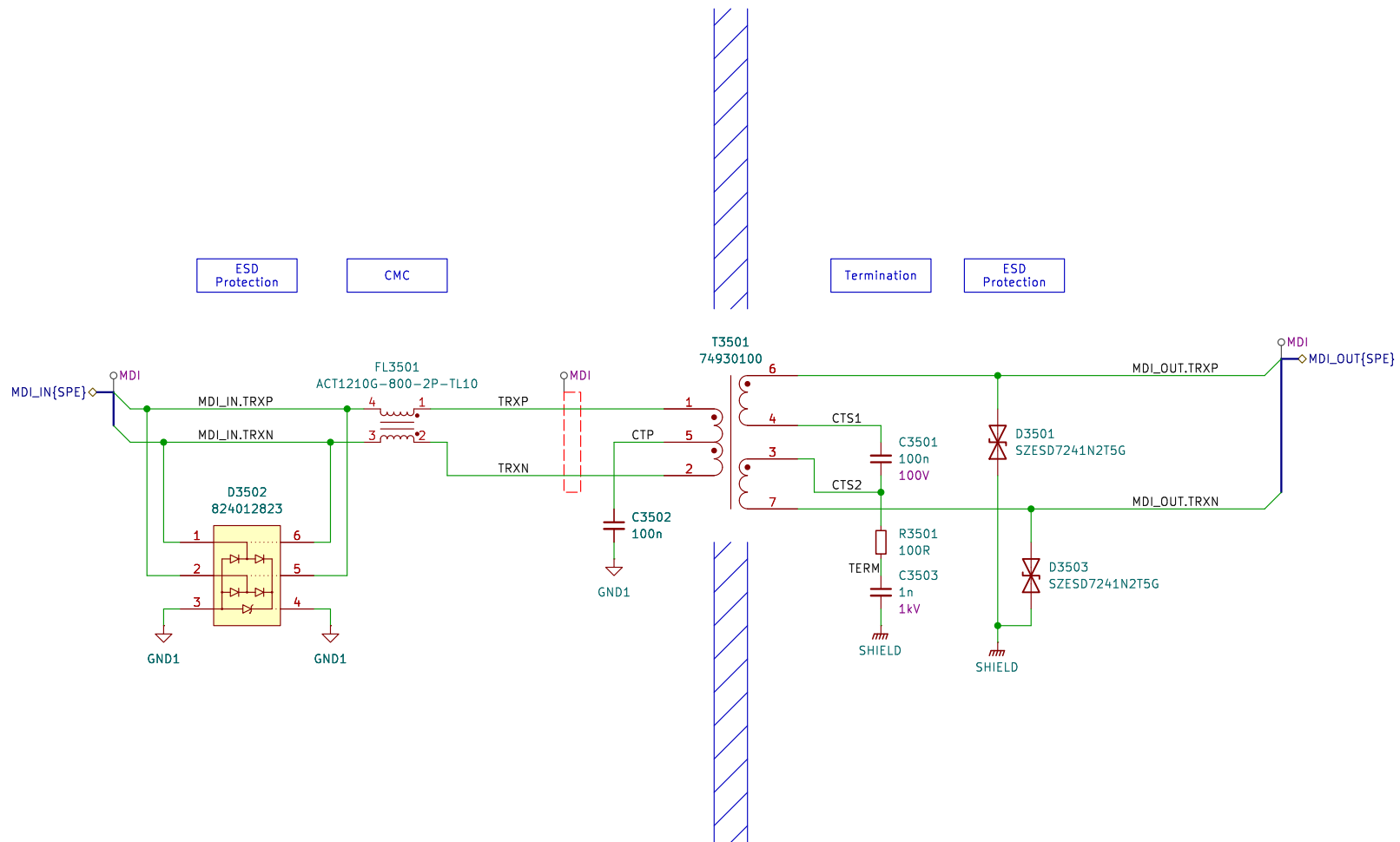
Size: A4
KiCad E.D.A. 9.0.6

Date:

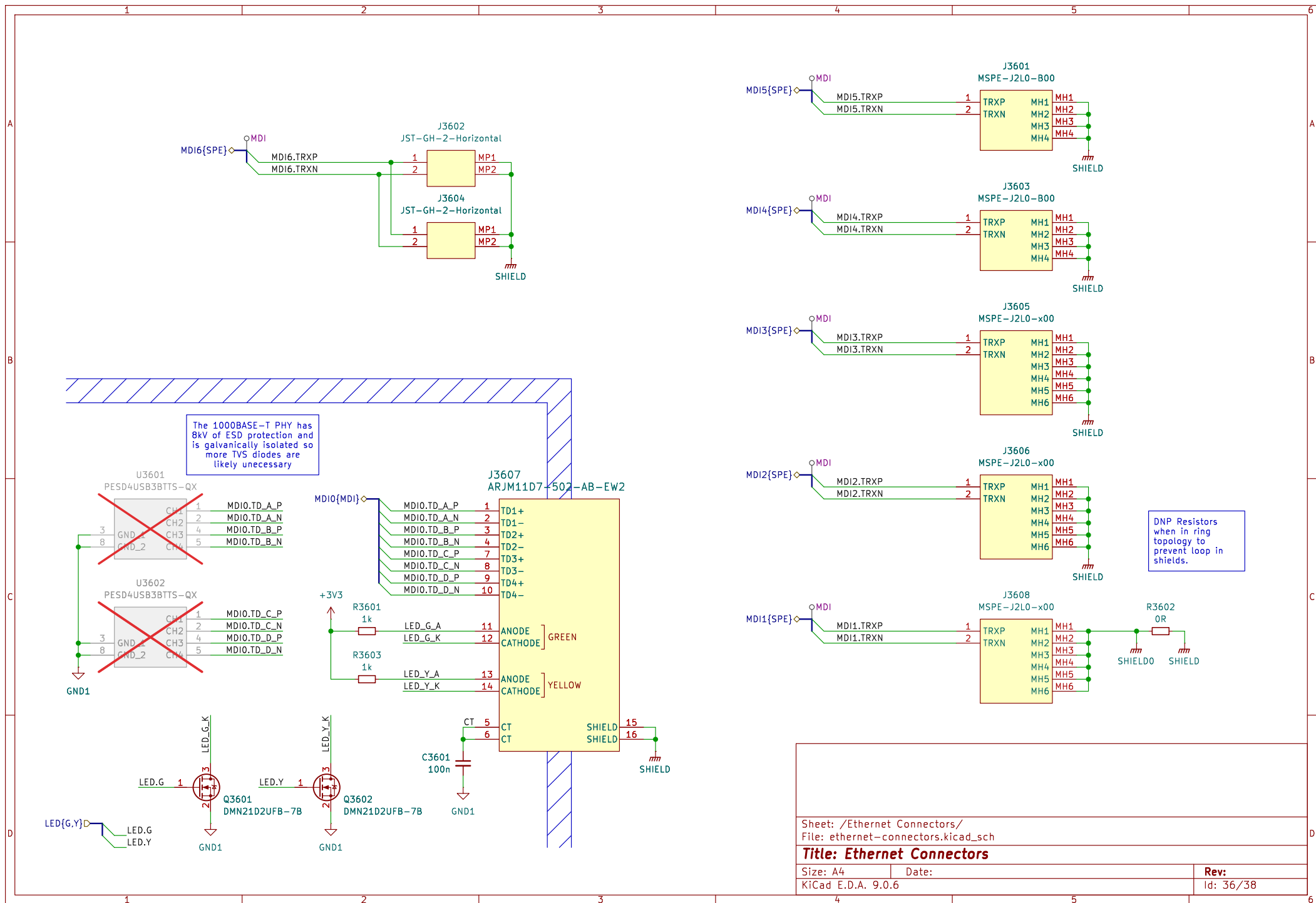
Rev:
Id: 33/38

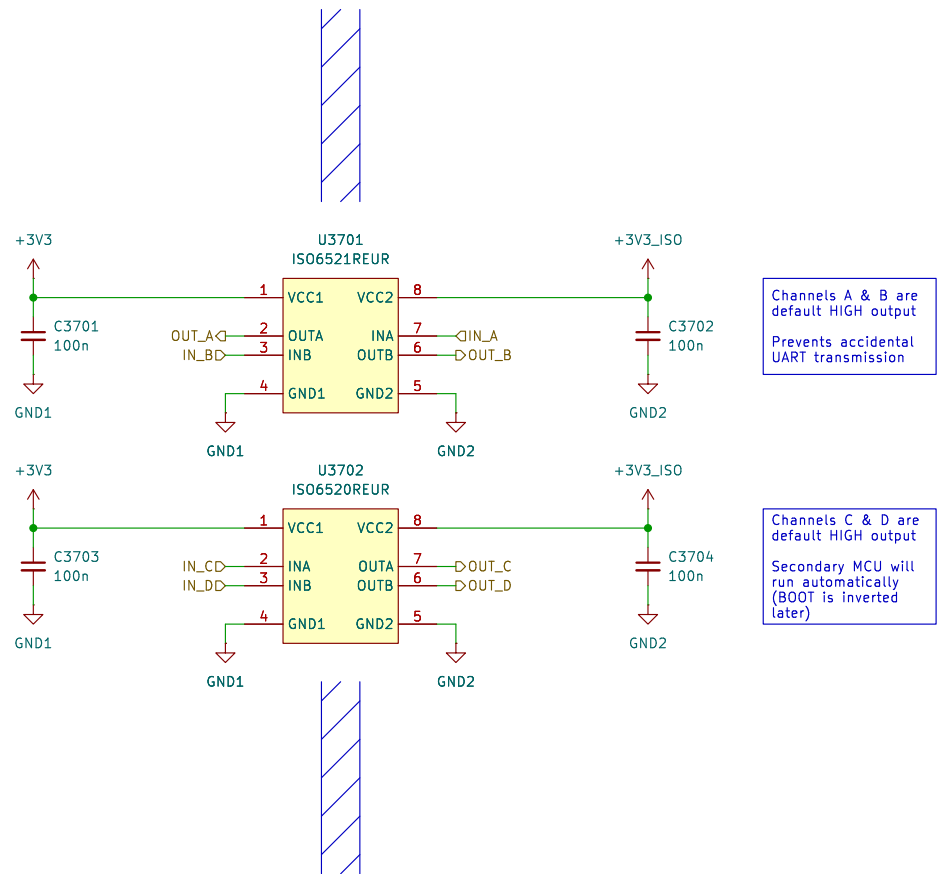


Coupling-decoupling network for 1000BASE-T1		
Sheet: /CDNs/1000BASE-T1 CDN 2/		
File: 1000base-t1-cdn.kicad_sch		
Title: 1000BASE-T1 CDN		
Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.6		Id: 34/38



Coupling-decoupling network for 1000BASE-T1		
Sheet: /CDNs/1000BASE-T1 CDN 3/		
File: 1000base-t1-cdn.kicad_sch		
Title: 1000BASE-T1 CDN		
Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.6		Id: 35/38





Sheet: /Isolation Crossing/ File: isolation-crossing.kicad_sch		
Title: Isolation Crossing		
Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.6		Id: 37/38

